

Derivative Supports of x^{x^a}

Leading asymptotics for every integer exponent and for the half exponent;
fixed-defect cancellations for rational, algebraic and irrational exponents

Built from seven research notes prepared for Vladimir Reshetnikov with OpenAI
(batch 72A manuscripts 28, 31, 34, 26, 20, 20r and 16)

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Abstract

For a positive exponent a , write the N th derivative of x^{x^a} as x^{x^a} times a finite sum of monomials $x^{a\ell-N}(\log x)^k$, and let $A_a(N)$ count the nonzero ones.

Integer exponents (Part I). For every fixed positive integer a we prove $A_a(N) = N^2/2 + o_a(N^2)$. An exact coefficient transformation gives positivity throughout the nonnegative power columns. In the remaining tail, circles in the coordinate $z = \tanh(t/2)$ and horizontal lines provide globally dominant saddles; a convex-envelope argument proves complete contour coverage; nonzero phase curvature, an integer-triangle estimate and compact exhaustion give density one of nonzero tail coefficients. Every negative-column coefficient at depths one and two is classified, and outside a countable algebraic set of real exponents the count is exactly $N(N+3)/2$. For $a = 1$ this is the asymptotic $a(N) \sim N^2/2$ of OEIS A293239, equivalently density zero for the zeros of the first-kind Lehmer–Comtet triangle; for $a = 2$ it is $\lim a(N)/N^2 = 1/2$ for OEIS A290268. Both were also proved separately, with more explicit detail at those exponents, and are printed as second routes.

The half exponent (Part II). For $x^{\sqrt{x}}$ the same leading asymptotic holds, by a circle-only saddle selection, and two infinite families of cancellations lie off the reflection line.

Fixed defects (Part III). On a fixed factor defect $r = N - \ell$: for rational $a \neq 1$ and $r \geq 4$ only finitely many coefficients vanish off an explicit reflection line (Schur, Siegel, and two certified quartics); for algebraic a of degree D the same holds on $r \geq 4D + 2$; for irrational a there are at most $(2\lfloor r/2 \rfloor)^2 - 1$ cancellation cells, effectively, at most one on each of defects two and three, and none for transcendental a ; for rational $1/2 < a < 1$ the nonreflection cancellations on defects two and three are counted exactly to $O(1)$ by explicit square-root laws.

All asymptotic statements are ineffective: no power-saving remainder or numerical order threshold is asserted, exact noncancellation at arbitrary depth is not proved, and no exponent grows with N . The proofs are conventional mathematics, not proof-assistant formalizations or refereed publications.

[merge] This report merges seven AI-assisted research notes dated 1 October 2026, all pinned to commit `ca81647a9` of the neighbouring report *Term counts in the derivatives of power towers* (`power-tower-derivative-term-counts`), except source 20, which names no pin. It answers that report’s Question 6 and the closing remark of its Part 2. Section 1 says what each note contributed and where the merge had to choose; Section 2 fixes the notation.

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1 How this report is built

1.1 Sources

The report is built from seven manuscripts of batch 72A of ProveIt’s incoming-reports intake. Batch 72 arrived in commit 1512ef835; the manuscripts are numbered by archive modification time as in the batch index, and their code and data were placed in commit d292c6765. Source 20r was never delivered on its own: it arrived only as `rational_companion_source.zip` (with its PDF) inside archive 20. Every source carries the author line “Research note prepared for Vladimir Reshetnikov with OpenAI”. The zip entry times show that the notes were written in the order 34, 31, 28, 26, 20r, 20, 16 on 1 October 2026, which is the order in which “earlier” and “preceding” in their texts should be read.

Table 1: The seven sources. “Pin” is the commit of the host report each source inspected; “pp.” is the length of the delivered PDF, which is not shipped.

Source	Archive and delivered title	Pin	Printed here
28 (base)	ProveIt_All_Integer_Exponent_Derivative_Supports; <i>Derivative Supports of x^a: a common leading asymptotic for every fixed positive integer exponent</i> , 10 pp.	ca81647a9	Sections 3–11
31	ProveIt_A293239_Leading_Asymptotic; <i>The Leading Asymptotic of OEIS A293239: sparse zeros of the Lehmer–Comtet triangle</i> , 8 pp.	ca81647a9	Section 12 (second route at $a = 1$)
34	ProveIt_A290268_Leading_Asymptotic; <i>The Leading Asymptotic of OEIS A290268</i> , 11 pp. (main file and <code>tanh_sections.tex</code>)	ca81647a9	Section 13 (second route at $a = 2$)
26	ProveIt_Half_Exponent_Derivative_Supports; <i>Derivative Supports of $x^{\sqrt{x}}$: a leading-density theorem with two infinite extra-cancellation families</i> , 7 pp.	ca81647a9	Section 14
20r	inside ProveIt_Algebraic_Exponents_and_Cancellation_Frequencies, as <code>rational_companion_source.zip</code> (folder <code>Rational_Exponent_Fixed_Defects</code>); <i>Reflection and Finite Exceptions: fixed defects in derivatives of x^a for positive rational $a \neq 1$</i> , 6 pp.	ca81647a9	Sections 15, 16
20	ProveIt_Algebraic_Exponents_and_Cancellation_Frequencies; <i>Algebraic Degrees and Exact Cancellation Frequencies: two fixed-defect extensions for derivatives of x^a</i> , 4 pp.	none named	Sections 15, 17
16	ProveIt_Irrational_Exponents_Effective_Cancellation_Bounds; <i>Effective Bounds for Irrational Exponents: fixed-defect cancellations in derivatives of x^a</i> , 6 pp.	ca81647a9	Sections 15, 18

Every result, proof, example, remark, question and limitation of the seven sources is printed. Section headings carry the source number; within a section the text is the source’s, apart from the renamings of Section 2 and the passages marked **[merge]**, which were written when the sources were merged. Each source’s closing section on verification, source position and scope is printed in Sections 19 and 20.

1.2 What was merged, and where the merge had to choose

No source cites another, except that 20 cites 20r as its companion. Apart from the coefficient setup shared by 20r, 20 and 16, the duplication between them is mathematical, not textual.

- **One theorem, three proofs.** Source 28 proves $A_a(N) = N^2/2 + o_a(N^2)$ for every fixed positive integer a (Theorem 3.1). Source 31 proved the case $a = 1$, OEIS A293239, and source 34 the case $a = 2$, OEIS A290268, half an hour and an hour earlier, by the same contour method (tanh circles, horizontal lines, the kernel $U(s, y)$, the integer-triangle lemma and compact exhaustion). The general theorem is stated once. The statements of 31 and 34 are printed as Corollaries 12.1 and 13.1, in the forms their sources gave them, each followed by its source's complete proof as a second route. Those proofs are kept because their steps are genuinely more explicit at the two exponents: source 31 classifies every critical point and identifies the real fold $e^{T_*} = 1 + T_* + T_*^2$, $\rho_* \approx 1.29843$; source 34 proves real-saddle positivity on the whole strict real region and gives a rational curvature certificate at $T = 14/5$. All three proofs are credited at each statement.
- **One lemma, four proofs.** The integer-triangle lemma is proved with the same argument in sources 28, 31, 34 and 26. It is printed once, as Lemma 9.1, whose proof includes the two details that only 31 and 34 spell out; the four labels of the sources all point to it.
- **One coefficient setup.** Sources 20r, 20 and 16 open with the same centered coefficient identity, in nearly the same words. It is printed once, in Section 15, in source 20r's words, together with the degree lemma that only source 16 contains.
- **A finiteness theorem implied twice.** Source 20's Theorem 17.1 (finitely many nonreflection cancellations on every fixed defect $r \geq 4D + 2$, for algebraic a of degree D) is implied by 20r's Theorem 16.1 for rational a and by 16's Theorem 18.1 for irrational a . Its proof, by field degrees and Siegel's theorem over $\mathbb{Q}(a)$, is a different route and is printed as such. Source 20's counting theorem and its window corollary are new and are printed in full.
- **One pair of families, two derivations.** Source 26's two infinite zero families at $a = 1/2$ (Theorem 14.3) are 20r's residual formulas (16.12) specialized to $a = 1/2$, and 20's endpoint remark counts them. The families are stated once, as Theorem 14.3 with 26's zero lists and Bell-block proof; the specialization is recorded where 20r's formulas are printed.
- **A clause implied by a corollary.** The transcendental clause of 16's Theorem 18.1 also follows from 28's Corollary 10.1.
- **Re-derivations of the host.** Several statements re-derive results already proved in **power-tower-derivative-term-counts**. They are kept, and the host statement is cited next to each, as a second route and never as new: 28's Proposition 3.2 at $a = 2$, Theorem 11.1 at $a = 1, 2$ and (11.1) at $a = 2$; 31's expansion and count; 34's coefficient model, its tanh coefficient formula and its upper bound, which is weaker than the host's.

The merge chose 28 as the base because it is the most general of the three proofs, and printed the $a=1$ and $a=2$ material after it, rather than splitting 31 and 34 into the host report, which would have printed the $a=1, 2$ theorem twice in the collection. It chose to print 20's algebraic-degree theorem in full although it is implied, because its proof is the only one of the three that works uniformly over $\mathbb{Q}(a)$. It renamed symbols only where two meanings would meet in one statement, and lists every renaming in Section 2.

1.3 Relation to the host report and to formal developments

The neighbouring report `power-tower-derivative-term-counts` (here called *the host*) studies the derivative supports of x^x (its Part 2, OEIS A293239), x^{x^2} (Part 3, OEIS A290268) and x^{x^x} (Part 4). Its numbering below is that of its text at the pin `ca81647a9`.

- **Answered: the closing remark of the host’s Part 2.** The host proves $n+1+\lfloor n^2/4 \rfloor \leq a(n) \leq q(n)$ for A293239 (its Theorem 2.16), shows that $a(n) \sim n^2/2$ is equivalent to $E(n) = o(n^2)$ and calls this “potentially a weaker target than the complete zero classification”, and closes Part 2 by noting that “it suffices to bound their cumulative number by $o(n^2)$ ” while its argument “proves quadratic order but not this sharper estimate”. Theorem 3.1 at $a = 1$, equivalently Corollary 12.1, proves that estimate. It is ineffective, so the host’s explicit lower bounds (its Theorem 2.16 and Corollary 2.17) remain the only explicit ones.
- **Answered: the host’s Question 6** (Section 3.24): “Establish the conjectured leading term $a(N) \sim N^2/2$ without first proving every individual nonvanishing claim.” Theorem 3.1 at $a = 2$, equivalently Corollary 13.1, does so. The explicit lower bounds of the host (leading constant $3/8$) are not superseded as explicit bounds.
- **Partly answered: the host’s Question 8** (extend the method to x^{x^r} , $r \geq 3$, after rederiving the depth generating function and the symmetry centre) and its Research question 3.68 (“which functions x^{x^m} have only symmetry-forced holes, and for which m do sporadic cancellations occur?”). The leading asymptotic for every integer exponent, the depth generating function (3.6), the symmetry centre $q = ad$ and the complete first two depths are here (Part I); for $a = 1/2$ there are infinite non-reflection families (Part II), and for non-integer rational, algebraic and irrational a each fixed defect has only finitely many non-reflection zeros (Part III). Exact noncancellation at arbitrary depth remains open for every exponent.
- **Bears on the host’s synthesis.** The host’s Section 5 observes that every cancellation it can prove in its three functions is a parity zero of one falling factorial about its centre. That remains true of those three functions. For $a = 1/2$ (Theorem 14.3) and for every rational $1/2 < a < 1$ (Corollary 16.3) there are infinite cancellation families that are not reflection zeros.

The host itself was not edited when this report was written. Its dated notes and reciprocal remarks pointing here are a separate change.

None of the results of this report is formalized in Lean or Rocq. The repository’s Lean library for A290268 (`Combinatorics/DerivativeExpansions/A290268/Lean`) defines the coefficients and the count of Corollary 13.1, `LeanProofs.A290268.coeff` and `LeanProofs.A290268.a`, in the same lattice $x^{x^2+j}(\log x)^k$ as Section 13. It proves the support window, the table identity `a_values_le_53` for $n \leq 53$, and the conditional reduction `a_eq_closedForm_of_support_all`. It contains no asymptotic statement, and nothing here discharges its hypotheses. No Lean or Rocq development exists for x^x , for the Lehmer–Comtet triangle, or for x^{x^a} with $a \neq 2$.

2 Conventions and notation

2.1 Coordinates

Throughout, $a > 0$ is the exponent (a positive integer in Part I, $a = 1/2$ in Part II, a positive real in Part III) and

$$\frac{d^N}{dx^N} x^{x^a} = x^{x^a} \sum_{\ell, k} p_{N, \ell, k}(a) x^{a\ell - N} (\log x)^k.$$

Here ℓ is the number of exponent factors and k the power of $\log x$. Table 2 lists the derived coordinates and how the host writes them. At an integer exponent, Part I writes the same coefficient as $c_{a, N}(j, k)$, the coefficient of $x^j (\log x)^k$ with $j = a\ell - N$.

Table 2: Coordinates used throughout, against the host report. “Host, $a = 1$ ” is the host’s Part 2 and “host, $a = 2$ ” its Part 3 (the depth-coordinate subsections 3.14–3.26 use the letters of this report).

Here	Meaning	Host	Tempting false reading
$d = \ell - k$	depth	$a = 1$: the column r of $b(m, r)$; $a = 2$: the deficit $m = k - j$	false: the host’s Part 2 $d = m - r$, an <i>offset</i>
$r = N - \ell$	factor defect (Parts II, III)	$a = 1$: $N - \ell$ is the offset d of Part 2; $a = 2$: $n - k$	false: the host’s $r = M - k$ of Section 3.14 or its column r
$q = -j - 1$	tail offset, $N = a(k + d) + q + 1$	$a = 2$: q of Section 3.14, which is $d = n - 2k - 1$ elsewhere in Part 3	false: the denominator of a rational a , which is written q here
$M = N - k$	coefficient index	$a = 2$: M of Section 3.14	—
$\kappa = k/d, \eta = q/d$	ratios (Part I)	not used	false: $\hat{\eta}$ of Part II, which is a different ratio
$b(M, d)$	Lehmer–Comtet number, $d = \text{column}$ (Section 12)	$b(m, r)$, $r = \text{column}$	false: d as the host’s offset
$H_{a; d, k, q}$	the integer (3.4); $H_{d, k, q} = H_{2; d, k, q}$	$H_{d, k, q}$ of Section 3.14	false: the host’s $H(u) = \text{arctanh}(u)/u$
K	reflection form (15.5) (Part III)	—	false: the host’s constants K_d
$U(s, y)$	partial-fraction kernel (4.3)	—	false: the host’s proposed count $U(n) = n^2/2 + n + 1$ (Part 3)
$L_a(t), L_\rho(t), L(t)$	actions of the contour integrals	—	false: the host’s $L = \log x$
z	$\tanh(t/2)$ (Parts I, II), a formal series variable (Part III)	x^2 in Part 3	false: $z = x^2$
e, e	Euler’s number; a tilt magnitude	—	false: e as Euler’s number
$P_d(\tau)$	$\text{Im}(\tau + i\pi)^d$ (Section 11)	$\pi R_d(\tau)$ in Prop. 3.30	false: a Gaussian prefactor P, P_c, P_h , or the host’s P_n

2.2 Renamed symbols

Where one letter carried two meanings that would meet in this report, one of them was renamed. No normalization was changed.

- The ratio $r = a + \eta$ of sources 28, 31 ($r = \rho$) and 34 ($r = \eta + 2$) is written ω , because r is the factor defect in Parts II and III. Source 34's critical map $F_{r,e}$ is $F_{\omega,e}$.
- Source 28's polynomial $R_a(x) = x^a - (-1)^a$ in Section 11 is $\mathcal{R}_a(x)$, because $R_{a,r,N}$ is the residual of Part III.
- In source 34: the functions $A(s)$, $B(s)$ of the horizontal lemma are $B_1(s)$, $B_2(s)$, as in source 28, which uses those names for the same functions; the circle parameter a and radius ρ of the tanh sections are α and R , as in source 28; the phase curvature $C(\kappa, \eta)$ is written $\text{Im } D^2 S$ with $D = \partial_\kappa - 2\partial_\eta$, source 28's notation at $a = 2$; the branch functions $K(T)$, $F(T)$ of the curvature seed are $\mathcal{K}(T)$, $\mathcal{F}(T)$.
- In source 26: the ratio $\eta = 2\nu + \kappa - 1$ is $\hat{\eta}$ (it is not source 28's $\eta = q/d$), and the favoured real point $z = -r$ of Section 14.3 is $z = -\sigma$, because r is the defect in Section 14.6.
- In Part III a rational exponent is written $a = \mathfrak{p}/\mathfrak{q}$ (sources 20r and 20 write p/q), because p is the coefficient $p_{N,\ell,k}$ and q the tail offset. Source 20's three uses of L become $\mathbb{L} = \mathbb{Q}(a)$, $\widetilde{K} = \mathfrak{q}K$ and the Laguerre polynomials $L_m^{(\pm 1/2)}$, which keep their name.
- In source 16: the leading line $L_a(N, k)$ is $\Lambda_a(N, k)$ (source 28's $L_a(t)$ is an action); in Section 18.3 the coefficient sum C_a is Σ_a (source 28's $C_a(z)$ is a polynomial), the constant $A_a = (C_a/|c_r|)^{1/e}$ is $\Gamma_a = (\Sigma_a/|c_r|)^{1/s_r}$ (source 16 also uses A_a for a power series) and the exponent $e = s_r$ is written s_r ; in the proof of Corollary 18.2 the local quantities $d = 2u - k$, $q = 1/u$, $t = d/u$ are δ , ϖ , τ . Source 16's $R_r(a; N, k)$ is $R_{a,r,N}(k)$ viewed in $\mathbb{Q}[a, N, k]$, and its K_a is K ; Section 15 uses the latter names, and Section 18 keeps source 16's.

2.3 Letters local to one section

The following letters keep their sources' meanings but only inside the section or proof where they are defined: $A(t)$, $B(t)$, $b(t)$, $E(T; \kappa)$, $M_a(\kappa)$ and W_a (Part I), where also α is the circle parameter $(1 - R^2)/(1 + R^2)$ in Sections 5 and 13, but a real exponent, with E an exceptional set, in Section 10; s , y , $C = \cosh s$, $S = \sinh s$, $c = \cos y$ and $D = s^2 + y^2$ in the contour estimates (where S is also, in the cosine-law subsections, the stationary action); the constants m , M , A , c and the block length H of Lemma 9.1; $G(t)$, $F(t)$, $J(s)$ and h in Section 12; ν , $E(\sigma; \kappa)$, $M(\kappa)$, $F(z, \varepsilon)$, $h = \sqrt{\varepsilon}$, $D = \partial_\kappa - \partial_\nu$ and $L = \log x$ (in Section 14.6) in Part II; and U , V , b , e , f_r , Y_r , C , C_r , E_r , $m = \lfloor r/2 \rfloor$ in Part III. The letter h always denotes a decay height $\min(a, \eta)$ in Part I contour estimates, with the values $\min(1, \rho - 1)$ at $a = 1$ and $\min(2, \eta)$ at $a = 2$.

2.4 Part numbers in the sources

Sources 31 and 28 call the host's parts "Part I" (for x^x) and "Part II" (for x^{x^2}), numbering only its three investigations. The host numbers them Part 2 (Section 2) and Part 3 (Section 3). Where those words are printed, a **[merge]** note gives the host's numbering.

Part I

Integer exponents

[merge] Sections 3–11 are source 28, the base of this report, in its own order. Section 12 is source 31 (the exponent $a = 1$) and Section 13 is source 34 (the exponent $a = 2$). Source 28's closing section is printed in Sections 19 and 20.

3 Statement and exact coefficient coordinates (source 28)

For $a \in \mathbb{Z}_{\geq 1}$, write

$$\frac{d^N}{dx^N} x^{x^a} = x^{x^a} \sum_{j \in \mathbb{Z}, k \geq 0} c_{a,N}(j, k) x^j (\log x)^k,$$

with like monomials collected, and let $A_a(N)$ count the nonzero coefficients.

Theorem 3.1. *For every fixed positive integer a ,*

$$A_a(N) = \frac{N^2}{2} + o_a(N^2) \quad (N \rightarrow \infty). \quad (3.1)$$

More precisely, only $o_a(N^2)$ candidate coefficients cancel in the negative-column tail specified below. The theorem does not classify individual extra zeros, provide a global power-saving remainder, or assert uniformity when a grows with N .

[merge] Theorem 3.1 is proved by source 28 for every fixed positive integer a (Sections 3–9). Its two cases that are OEIS sequences were proved separately and earlier on the same day, by the same contour method: $a = 1$, the derivatives of x^x (OEIS A293239), by source 31, printed as Corollary 12.1 with its proof in Section 12; and $a = 2$, the derivatives of x^{x^2} (OEIS A290268), by source 34, printed as Corollary 13.1 with its proof in Section 13. None of the three sources cites the others. At $a = 1$ the theorem proves the estimate $E(n) = o(n^2)$ that the host report’s Part 2 leaves open in its closing remark; at $a = 2$ it answers the host’s Question 6 (Section 1.3). Like those two proofs, it is ineffective.

Taylor expansion at a fixed $x > 0$ gives

$$c_{a,N}(j, k) = \frac{N!}{k!d!} [w^N] ((1+w)^a - 1)^k (1+w)^{ad} \log^d(1+w), \quad j = a(k+d) - N. \quad (3.2)$$

Indeed the logarithm of the Taylor ratio splits into $x^a((1+w)^a - 1) \log x + x^a(1+w)^a \log(1+w)$. Set

$$\begin{aligned} q &= -j - 1, & N &= a(k+d) + q + 1, \\ M &= N - k = (a-1)k + ad + q + 1, \end{aligned} \quad (3.3)$$

$$H_{a;d,k,q} = \frac{M!}{d!} [w^M] \left(\frac{(1+w)^a - 1}{w} \right)^k (1+w)^{ad} \log^d(1+w). \quad (3.4)$$

Then $c_{a,N}(-q-1, k) = \binom{N}{k} H_{a;d,k,q}$.

Define the even polynomial

$$C_a(z) = \frac{(1+z)^a - (1-z)^a}{2az} = \sum_{m \geq 0} \frac{1}{a} \binom{a}{2m+1} z^{2m}. \quad (3.5)$$

It has positive coefficients on its even support and constant term one. The two changes of variable $w = e^t - 1$ and $z = \tanh(t/2)$ yield the exact identity

$$H_{a;d,k,q} = \frac{M!}{d!} a^k 2^{-(a-1)(k+d)-q-1} [z^M] C_a(z)^k (\operatorname{arctanh} z)^d (1+z)^{ad} (1-z)^q. \quad (3.6)$$

This algebraic identity is valid also for negative q whenever the original indices are admissible. Its powers include the transformed Cauchy differential.

[merge] At $a = 2$ one has $C_2 = 1$, and (3.6) is source 34’s identity (13.16), which is in turn the host’s transformed coefficient formula (its Proposition 3.6, `xxb:prop:mobius`) written in depth coordinates. At $a = 1$ it is source 31’s formula (12.17), with $H_{1;d,k,q} = b(M, d)$.

Proposition 3.2 (The exact positive bulk). *Every candidate with $j \geq 0$ is strictly positive. The $j = -1$ boundary is strictly positive when $d \geq 1$ and structurally zero when $d = 0$.*

Proof. For $j \geq 0$, one has $q \leq -1$. Every factor in (3.6) has nonnegative coefficients, and $(1 - z)^q$ has strictly positive coefficients at every degree. Since $M = N - k \geq d$, the extracted coefficient is positive. At $q = 0$, $d \geq 1$, the series $(\operatorname{arctanh} z)^d$ supplies every degree of its parity above d , and $(1 + z)^{ad}$ supplies both parities. At $d = 0$, $q = 0$, the index $M = (a - 1)k + 1$ exceeds the degree of C_a^k , so the coefficient is zero. $\square$

The recurrence obtained by differentiating once preserves

$$\ell = k + d = (N + j)/a \in \mathbb{Z}, \quad 0 \leq k \leq \ell \leq N.$$

Differentiation leaves ℓ fixed and leaves k fixed or lowers it by one; multiplication by $x^{a-1}(a \log x + 1)$ raises ℓ by one and raises k by zero or one. Thus $A_a(N) \leq (N + 1)(N + 2)/2$. The positive nonnegative columns contain

$$\sum_{\ell=\lceil N/a \rceil}^N (\ell + 1) = \left(\frac{1}{2} - \frac{1}{2a^2} \right) N^2 + O_a(N). \quad (3.7)$$

The outside tail $d \geq 1$, $k \geq 0$, $q \geq 1$ has

$$\sum_{\ell=1}^{\lfloor (N-2)/a \rfloor} \ell = \frac{N^2}{2a^2} + O_a(N) \quad (3.8)$$

candidates. We prove density one of nonzero coefficients there.

[merge] At $a = 2$, Proposition 3.2 is the host's nonvanishing region (its Theorem 3.7, `xxb:thm:positive`), whose top logarithmic column is its Proposition 3.9, and the bulk count (3.7) has the leading term $3N^2/8$ of the host's exact lower bound $\Lambda(N) = \beta(N)$ (its Theorem 3.1 and equation (3.85)); source 34 cites the same result as (13.4). The upper bound $(N + 1)(N + 2)/2$ is source 34's (13.6) at $a = 2$; there the host's Theorem 3.14 proves the sharper $a(N) \leq U(N) = N^2/2 + N + 1$.

4 A meromorphic action (source 28)

For now let $d \geq 1$, $\kappa = k/d > 0$, $\eta = q/d > 0$, and $\eta \neq a$. Put

$$\omega = a + \eta, \quad e = \frac{|a - \eta|}{2}, \quad h = \min(a, \eta) > 0.$$

Here e denotes a tilt magnitude and e denotes Euler's number. Formula (3.4) becomes

$$H_{a;d,k,q} = \frac{M!}{d!} \frac{1}{2\pi i} \oint \frac{\exp(dL_a(t))}{4 \sinh^2(t/2)} dt, \quad (4.1)$$

where

$$\begin{aligned} L_a(t) &= \log t + \frac{a - \eta}{2} t + \kappa \log \sinh(at/2) - (a\kappa + a + \eta) \log \sinh(t/2) \\ &\quad - ((a - 1)\kappa + a + \eta) \log 2 \\ &= \log t + at + \kappa \log(e^{at} - 1) - (a\kappa + a + \eta) \log(e^t - 1). \end{aligned} \quad (4.2)$$

The integer powers make the full integrand single-valued. Its nonzero poles are at $2\pi i\mathbb{Z}$; numerator zeros are zeros of the integrand, not singularities. The residual differential under $z = \tanh(t/2)$ is exactly $dz/(2z^2)$, proving (3.6).

Reflect the real coordinate if $\eta < a$, so the favored side is left. The reflected modulus is obtained by replacing $(a - \eta)t/2$ by $-et$; call its action L_c . For $t = -s - iy$, $s > 0$, $0 < y < \pi$, define

$$U(s, y) = \frac{s \coth s - y \cot y}{s^2 + y^2}, \quad V(s) = \frac{\omega}{2} - e \coth s. \quad (4.3)$$

The partial fractions of $\coth$ and $\cot$ [6] give

$$U(s, y) = 2 \sum_{n \geq 1} \frac{(n\pi)^2}{(s^2 + (n\pi)^2)((n\pi)^2 - y^2)}.$$

Consequently $U_s < 0$, $U_y > 0$, and $V' > 0$.

5 A uniform angular-selection mechanism (source 28)

5.1 Tanh circles, including circles through numerator zeros

The roots of C_a are purely imaginary:

$$C_a(z) = \prod_{j=1}^{\lfloor (a-1)/2 \rfloor} (1 + z^2/\lambda_j), \quad \lambda_j = \tan^2(\pi j/a). \quad (5.1)$$

An empty product equals one. On $|z| = R < 1$, let $\alpha = (1 - R^2)/(1 + R^2)$. In the favored quadrant the image in the t plane satisfies

$$\cos y = \alpha \cosh s, \quad y' = -\tanh s \cot y < 0.$$

Write $C = \cosh s$, $v_j = R^2/\lambda_j$. The exact geometry yields

$$\begin{aligned} \cos^2(\arg z) &= \frac{1 - C^{-2}}{1 - \alpha^2}, \\ |1 + z^2/\lambda_j|^2 &= A_j - B_j/C^2, \quad B_j = \frac{4v_j}{1 - \alpha^2} > 0, \quad A_j = (1 - v_j)^2 + B_j \geq B_j. \end{aligned} \quad (5.2)$$

Hence

$$\frac{d}{ds} \log |1 + z^2/\lambda_j| = \tanh s \frac{B_j}{A_j C^2 - B_j}.$$

The other κ -dependent power of $|z|$ is constant on the circle. Direct differentiation of the full action gives

$$\frac{d}{ds} \operatorname{Re} L_c(-s - iy(s)) = \tanh s \left[U(s, y(s)) - V(s) + \kappa \sum_j \frac{B_j}{A_j C^2 - B_j} \right]. \quad (5.3)$$

Every term in the bracket decreases strictly in s : $U_s + U_y y' < 0$, $-V' < 0$, and each rational summand decreases. The derivative is positive near the imaginary endpoint. If $R^2 = \lambda_j$, the logarithm tends to $-\infty$ at that endpoint and its derivative diverges positively; the maximum remains away from the zero. A zero-weight factor when $\kappa = 0$ is simply omitted.

It follows that each favored quadrant has one maximum, either a strict interior maximum or its real endpoint. Reflection to the other half-contour lowers the real action by $2e|\operatorname{Re} t|$. In particular every off-axis saddle of height less than $\pi/2$ is globally selected on its tanh circle, with nonzero angular curvature. This includes $a \geq 5$, when some numerator zeros lie inside the unit disk.

5.2 Horizontal lines and a finite Jensen inequality

Fix $\pi/2 \leq y \leq \pi$ and write $C = \cosh s$, $S = \sinh s$, $c = \cos y \leq 0$ and $D = s^2 + y^2$. For $u(s, y) = \operatorname{Re} L_c(-s - iy)$, the multiplication formula for coth gives

$$\begin{aligned} u_s \frac{C - c}{S} &= B_1(s) + eB_2(s) - \frac{\omega}{2} + \kappa W_a(C, c), \\ B_1 &= \frac{s(C - c)}{DS}, \quad B_2 = \frac{C - c}{S}, \\ W_a(C, c) &= \frac{1}{2} \sum_{j=0}^{a-1} \frac{c_j - c}{C - c_j}, \quad c_j = \cos(y + 2\pi j/a). \end{aligned} \tag{5.4}$$

Both B_1 and B_2 strictly decrease, because

$$B_2' = (Cc - 1)/S^2 < 0, \quad \frac{1}{B_1} = 2 \sum_{n \in \mathbb{Z}} \frac{s^2 + y^2}{s^2 + (y + 2\pi n)^2}.$$

All summands in the second identity are nondecreasing and some strictly increase; the differentiated series converges locally uniformly.

For $a \geq 2$, the numbers c_j average to zero. With

$$g(x) = \frac{x - c}{(C - x)^2}, \quad g''(x) = \frac{4C + 2x - 6c}{(C - x)^4} > 0 \quad (-1 \leq x \leq 1),$$

Jensen's inequality gives $\sum_j g(c_j) \geq ag(0) = -ac/C^2 \geq 0$. Thus W_a is nonincreasing in C . Equality throughout is possible at $a = 2$, $y = \pi/2$, but the other terms still decrease strictly. For $a = 1$, $W_a = 0$ identically, so the same conclusion holds.

The bracket in (5.4) runs from $+\infty$ to $-h$. It therefore gives a unique favored-side maximum $s(y) > 0$, with $u_{ss} < 0$. Reflection strictly suppresses the opposite side. The resulting horizontal envelope $J(y) = u(s(y), y)$ satisfies

$$J'' = u_{yy} - u_{sy}^2/u_{ss} = -u_{ss} - u_{sy}^2/u_{ss} > 0, \tag{5.5}$$

by harmonicity at the maximum. At $y = \pi$ all the hyperbolic logarithmic factors have zero y derivative for every integer a , and therefore

$$J'(\pi) = \frac{\pi}{s(\pi)^2 + \pi^2} > 0. \tag{5.6}$$

The horizontal tails satisfy $\operatorname{Re} L_c \leq \log(1 + s) - hs + O(1)$. Closing the strip encloses only the original pole at zero, and the vertical contributions vanish. Thus every high favored saddle is globally selected. These assertions are locally uniform for $\kappa > 0$, $\eta > 0$, $\eta \neq a$.

6 Complete coverage from two convex envelopes (source 28)

Let $\mathcal{M}(\xi)$ be the maximum of $\operatorname{Re} L_a$ on $|z| = e^\xi$, $\xi < 0$. Up to a real additive constant, the modulus is

$$\log |\operatorname{arctanh} z| + \kappa \log |C_a(z)| + a \log |1 + z| + \eta \log |1 - z| - \nu \log |z|, \quad \nu = (a - 1)\kappa + a + \eta. \tag{6.1}$$

It is subharmonic on every annulus. Comparing it with a harmonic affine function of $\log |z|$ whose boundary values are the two boundary maxima proves that $\mathcal{M}$ is convex in ξ . Its slope near $-\infty$ is $1 - \nu < 0$.

The unique favored maximum in (5.3) makes the envelope differentiable, including a transition between a real endpoint and a conjugate pair. Conjugate maxima have the same radial derivative. At an interior complex maximum, harmonicity and its strict angular curvature give strictly positive envelope second derivative. The same holds at a nondegenerate real maximum. A flat interval with $\mathcal{M}' = 0$ is impossible: an interior maximum would have positive second derivative, while an interval of real stationary maxima would force L'_a to vanish identically, contrary to its pole at zero. Therefore a stationary minimum, if present, is unique.

As $\xi \uparrow 0$, the mapped circles tend locally to $y = \pi/2$. Their maxima remain in a bounded s interval: the uniform estimate $\log(1+s) - hs + O(1)$ makes large s unfavorable, whereas a fixed interior point has finite action. The maxima converge to the unique horizontal maximum $s(\pi/2)$. Since $dt/d\xi = \sinh t = -i \cosh s$ at the joining line,

$$\mathcal{M}'(0-) = \cosh(s(\pi/2))J'(\pi/2). \quad (6.2)$$

If this value is positive, convexity and the negative initial slope give a stationary circle with $R < 1$. If it is negative, (5.5) and (5.6) give a stationary horizontal line with $\pi/2 < y < \pi$. If it is zero, the joining contour itself selects the saddle. The selected critical point is unique up to conjugation.

Strict global gaps and nonzero curvature persist under small parameter changes. At the artificial joining height the contour can be perturbed locally, so it causes no genuine singularity of the selected simple saddle. We have proved complete selection of one real saddle or one complex conjugate pair, except for possible degenerate real saddles. This argument uses exact Cauchy cycles throughout.

7 The real boundary and connected oscillatory regions (source 28)

Use the original coordinate and put

$$A(t) = \frac{e^t}{e^t - 1}, \quad B(t) = \frac{ae^{at}}{e^{at} - 1} - aA(t), \quad b(t) = \frac{B(t)}{A(t)} = -a \frac{e^{(a-1)t} - 1}{e^{at} - 1}.$$

The positive real saddle equation is $\eta = E(T; \kappa)$, where

$$E(T; \kappa) = (a + 1/T)(1 - e^{-T}) - a + \kappa b(T). \quad (7.1)$$

For $a \geq 2$, it tends to $1 - a - (a - 1)\kappa < 0$ as $T \downarrow 0$, is positive for sufficiently large T , and tends to zero at infinity. Also $b(T) < 0$ and

$$E(T; \kappa) < \frac{1 - e^{-T}}{T} < 1.$$

Thus

$$M_a(\kappa) = \max_{T>0} E(T; \kappa) \quad (7.2)$$

is positive, less than one, continuous, and nonincreasing. For continuity, on each bounded κ interval a positive common lower bound for the maximum and the bound $E \leq 1/T$ confine all maximizers to a common compact interval.

When $\eta < M_a(\kappa)$, the first simple real crossing has $E_T > 0$, so $L_{tt} = AE_T > 0$ and its tanh angular curvature is positive. The angular-selection lemma makes it the selected global real maximum. When $\eta > M_a(\kappa)$, there is no positive real saddle. Negative real saddles are impossible at $\eta > 0$: for $t = -s$,

$$E(-s; \kappa) = \frac{e^s - 1}{s} - ae^s + \kappa b(-s) < 0.$$

Consequently the selected saddle is complex in the epigraph $\eta > M_a(\kappa)$. The two regions separated by $\eta = a$ are connected: below a , move through a horizontal level in $(1, a)$, and above a use the whole half-plane.

For $a \geq 2$ the real-degenerate parameter set has area zero. At a real $t \neq 0$, the two equations $L_t = L_{tt} = 0$ form a nonsingular affine system for (κ, η) , with determinant, up to sign,

$$A(t)^2 b'(t) > 0. \quad (7.3)$$

Indeed, for $x = e^t > 0$,

$$\frac{x^{a-1} - 1}{x^a - 1} = 1 - \frac{x^{a-1}}{1 + x + \dots + x^{a-1}},$$

and the final fraction strictly increases. The degenerate set lies in the image of a real-analytic map of one variable, hence is a countable union of compact curve images and has planar measure zero. Nongeneric real crossings cannot create another open complex region: neighboring simple first crossings are selected real, whereas a selected complex saddle would persist on an open neighborhood.

The endpoint $a = 1$. Here $b = 0$ and E is independent of κ . Writing $\rho = 1 + \eta$, its real equation is $(1 + 1/T)(1 - e^{-T}) = \rho$. Its unique turning point solves $e^T = 1 + T + T^2$, as follows directly by differentiating. The derivative changes sign once, so the real-fold set is a single line $\eta = M_1 \in (0, 1)$. The same first-real-root and complex-epigraph description follows. Both complex regions separated by $\eta = 1$ are connected. Thus the zero-area and connectedness conclusions hold for this endpoint as well.

[merge] Source 31 treats this endpoint in full (Section 12.2): with $\rho = M/d$, its saddle map is $F(t) = (1 + 1/t)(1 - e^{-t})$, the turning point is $T_* \in (1, 2)$ of (12.13), numerically $T_* = 1.79328\dots$, and the fold value is $\rho_* = F(T_*) = 1.29843\dots$, so that $M_1 = \rho_* - 1$. The values are those of the delivered diagnostic data/31-a293239-saddle_normalization.json.

8 Uniform local asymptotics and a nonzero curvature anchor (source 28)

Return to original coordinates. Let t_s be the selected lower complex saddle and set $S = L_a(t_s)$. On a tanh circle define

$$z_s = \tanh(t_s/2), \quad C_s = \sinh^2(t_s) L_{tt}(t_s), \quad P_c = \frac{1}{2z_s \sqrt{C_s}}. \quad (8.1)$$

The residual $dz/(2z^2)$ explains the factor $1/(2z_s)$. On a horizontal contour the equivalent lower-saddle factor is

$$P_h = -\frac{i}{4 \sinh^2(t_s/2) \sqrt{-L_{tt}(t_s)}}.$$

Square-root branches are fixed by the oriented Gaussian contour; the horizontal lower line runs left to right.

Proposition 8.1. *On every compact simple complex-saddle patch,*

$$H_{a;d,k,q} = \frac{2M!}{d!} \frac{e^{d \operatorname{Re} S} |P|}{\sqrt{2\pi d}} \left\{ \cos(d \operatorname{Im} S + \arg P) + O_a(d^{-1}) \right\}. \quad (8.2)$$

The error is additive and compact-uniform, including at zeros of the cosine. A selected real saddle has a single positive leading contribution and is eventually nonzero on each compact simple real-saddle patch.

Proof. The preceding sections supply an exact contour with a strict global gap and nonzero Gaussian curvature. In a local saddle coordinate, Taylor expansion under the $d^{-1/2}$ rescaling gives the Gaussian integral; odd terms cancel and the first remaining relative error is $O(d^{-1})$ for one saddle. The complementary contour is exponentially smaller. The conjugate contributions give (8.2), without division by its cosine. At a positive real saddle all leading factors are positive. $\square$

For a fixed- N row with d fixed, $\eta = (N-1)/d - a\kappa - a$. Set $D = \partial_\kappa - a\partial_\eta$. Affine parameter dependence and stationarity yield

$$DS = \log(e^{at_s} - 1), \quad D^2S = -\frac{[ae^{at_s}/(e^{at_s} - 1)]^2}{L_{tt}(t_s)}. \quad (8.3)$$

This phase curvature is not identically zero on either connected complex region. At $\kappa = 0$, $\eta = a$, the lower saddle is $t = -iT$, with

$$\tan(T/2) = aT, \quad \pi/2 < T < \pi, \quad L_{tt} = \frac{2a - 1 - a^2T^2}{2aT^2} < 0. \quad (8.4)$$

Its implicit η derivative has real part $1/(2L_{tt}) < 0$, identifying both favored continuations. For $a \geq 2$, at $T_0 = \pi(1 - 1/a)$ one has

$$\tan(T_0/2) = \cot(\pi/(2a)) < 2a/\pi < aT_0.$$

Strict monotonicity of $\tan(T/2)/T$ implies $T > T_0$. Hence $\delta = a(\pi - T)/2 \in (0, \pi/2)$, and

$$\cot(aT/2) = \begin{cases} -\cot \delta, & a \text{ even,} \\ \tan \delta, & a \text{ odd.} \end{cases}$$

Both values are finite and nonzero. For $a = 1$, $\cot(T/2) = 1/T > 0$ directly. Therefore

$$\text{Im } D^2S = -\frac{a^2 \cot(aT/2)}{2L_{tt}} \neq 0. \quad (8.5)$$

The anchor also excludes a numerator zero at $t = -iT$, so it persists for small positive κ . Connectedness and real analyticity now show that the curvature-zero set has area zero on both complex regions. Logarithm branches add only affine parameter terms and do not change this curvature.

[merge] At $a = 1$ the anchor (8.4) is source 31's centre (12.14) with the sign (12.15), and its curvature is the value $S''(2)$ computed after (12.25). At $a = 2$, source 34 uses a different, rational base point, $T = 14/5$ on the line $\eta = 2$ with $\kappa = \mathcal{K}(T) > 1/250$ (Section 13.7).

9 Integer points and full tail density (source 28)

We state the elementary lattice estimate with its proof.

Lemma 9.1 (Integer-triangle estimate; sources 28, 31, 34, 26). *Let $f \in C^2(J)$, where J has length $O(d)$, and suppose f'' has one sign with $m/d \leq |f''| \leq M/d$. For fixed $A, m, M > 0$, only $O(d^{2/3})$ integers $k \in J$ satisfy $\text{dist}(f(k), \mathbb{Z}) \leq A/d$.*

Proof. Split J into blocks of length $H = cd^{1/3}$. Round candidate values to integers within A/d . The integer determinant of any three rounded points in a block has absolute value at most $MH^3/(2d) + 4AH/d < 1$ for small fixed c and large d , so it is zero. All candidates in the block

are collinear. After changing sign, assume $f'' \geq m/d$. For a middle candidate x_i between the extreme candidates x_1, x_n , the chord inequality gives

$$\frac{m}{2d}(x_i - x_1)(x_n - x_i) \leq \frac{2A}{d}.$$

Integer spacing implies $\lfloor (n-1)^2/4 \rfloor \leq 4A/m$, hence $n \leq 2 + 4\sqrt{A/m}$. Each block contains only a constant number, and there are $O(d^{2/3})$ blocks. $\square$

[merge] Sources 31 (its Lemma 5.1), 34 (its Lemma 7.1) and 26 (its Lemma 5.1) state and prove the same lemma by the same argument, each for its own phase function. Two of them spell out steps that the proof above leaves implicit. Source 34 fixes the block constant by $Mc^3/2 < 1/4$. Source 31 makes the integer spacing explicit: for distinct integer abscissae and a middle index i , $x_i - x_1 \geq i - 1$ and $x_n - x_i \geq n - i$, which gives $\lfloor (n-1)^2/4 \rfloor \leq 4A/m$; and finitely many small d are absorbed in the implied constant. The applications in Sections 12, 13 and 14 cite this lemma. The letters m , M , A , c and H are local to the lemma: here M is not the coefficient index and H is not $H_{a;d,k,q}$.

On a compact complex patch where $|\operatorname{Im} D^2 S|$ is bounded below, an exact zero in (8.2) satisfies

$$\operatorname{dist}\left(\frac{d \operatorname{Im} S + \arg P}{\pi} - \frac{1}{2}, \mathbb{Z}\right) \leq \frac{C}{d}.$$

The phase second derivative with respect to integer k , at fixed d, N , is

$$\frac{\operatorname{Im} D^2 S}{\pi d} + O(d^{-2}).$$

Lemma 9.1 gives $O(d^{2/3})$ zero cells per row and $O(N^{5/3})$ on each such compact patch. Real-saddle patches contain no zeros for sufficiently large d .

For exhaustion, first restrict to compact physical rectangles bounded away from $\kappa = 0$ and $\eta = 0$, and remove a narrow neighborhood of $\eta = a$. Every positive real saddle there has $T \leq 1/\eta$ and is bounded away from zero. Thus its real-degenerate locus lies in a compact analytic-curve image. Remove a small-area neighborhood of this locus before taking compact analytic exceptional-set covers for the phase curvature. The remaining compact sets admit finite simple-saddle covers and the preceding bound. A finite union of boxes of arbitrarily small area covers the omitted sets; its lattice count is its scaled area plus $O(N)$. Taking $N \rightarrow \infty$ first and the removed area to zero afterward proves zero density on every compact physical rectangle.

The exact ratio-to-lattice map is

$$\left(\frac{d}{N-1}, \frac{k}{N-1}\right) = \left(\frac{1}{a(\kappa+1) + \eta}, \frac{\kappa}{a(\kappa+1) + \eta}\right),$$

with Jacobian $[a(\kappa+1) + \eta]^{-3}$. Its full integral is

$$\int_0^\infty \int_0^\infty \frac{d\eta d\kappa}{[a(\kappa+1) + \eta]^3} = \frac{1}{2a^2}. \quad (9.1)$$

This equals the leading candidate area in (3.8). Compact exhaustion therefore leaves an arbitrarily small normalized number of omitted candidates. The whole tail has only $o_a(N^2)$ zeros. Adding (3.7) and using the candidate upper bound proves Theorem 3.1.

10 An elementary exact count for generic real parameters (source 28)

The integer-exponent theorem contrasts with a simple algebraic genericity statement. Allow a real parameter $\alpha \neq 0$ and write

$$\frac{d^N}{dx^N} x^{x^\alpha} = x^{x^\alpha} \sum_{\ell, k} p_{N, \ell, k}(\alpha) x^{\alpha \ell - N} (\log x)^k.$$

Differentiation gives the polynomial recurrence

$$\begin{aligned} p_{N+1,\ell,k}(\alpha) &= (\alpha\ell - N)p_{N,\ell,k}(\alpha) + (k+1)p_{N,\ell,k+1}(\alpha) \\ &\quad + p_{N,\ell-1,k}(\alpha) + \alpha p_{N,\ell-1,k-1}(\alpha), \end{aligned} \quad (10.1)$$

starting from $p_{0,0,0} = 1$ and with out-of-range entries zero. Thus every coefficient belongs to $\mathbb{Z}[\alpha]$. For $N \geq 1$, the potential cells are $1 \leq \ell \leq N$, $0 \leq k \leq \ell$. None of these coefficient polynomials is identically zero: choose an integer $a > N$, for which every such cell lies in the strictly positive bulk of Proposition 3.2.

Corollary 10.1. *There is a countable set E of real algebraic numbers such that every real $\alpha \notin E$ has, for every $N \geq 1$, exactly*

$$\frac{N(N+3)}{2}$$

nonzero derivative monomials. In particular this exact count holds for every transcendental real α .

Proof. Take E to contain zero and all roots of the nonzero integer polynomials $p_{N,\ell,k}$. It is countable and algebraic. For $\alpha \neq 0$, distinct (ℓ, k) give distinct monomials: after $x = e^t$, finite sums of the functions $e^{(\alpha\ell - N)t} t^k$ with distinct exponents are linearly independent. All candidate coefficients are nonzero outside E , and their number is $\sum_{\ell=1}^N (\ell+1) = N(N+3)/2$. $\square$

This corollary does not classify the exceptional algebraic parameters. Such exceptions occur even at order two, where $p_{2,1,0}(\alpha) = 2\alpha - 1$ and $p_{2,1,1}(\alpha) = \alpha(\alpha - 1)$. Positive noninteger algebraic parameters therefore remain a meaningful separate frontier.

[merge] That frontier is the subject of Part III, on fixed defects $r = N - \ell$. Both exceptions displayed above lie on defect one, where the residual is $R_{\alpha,1,N} = -K/2$ (equation (16.12)): at $N = 2$ the reflection form (15.5) is $K = 1 - 2\alpha$ for $k = 0$ and $K = 1 - \alpha$ for $k = 1$, so $p_{2,1,0}$ vanishes at $\alpha = 1/2$ and $p_{2,1,1}$ at $\alpha = 1$, both on the reflection line. Corollary 10.1 implies the transcendental clause of source 16's Theorem 18.1, which source 16 proves per defect from the positive leading coefficient (15.8). In this section α is a real exponent and E an exceptional set; in Sections 5 and 7 the same letters denote the circle parameter $(1 - R^2)/(1 + R^2)$ and the real saddle map.

11 An exact classification at the first two depths (source 28)

The branch cut of the original coefficient formula gives an additional exact result. For $d \geq 1$, $k, q \geq 0$, set

$$\mathcal{R}_a(x) = x^a - (-1)^a, \quad P_d(\tau) = \text{Im}(\tau + i\pi)^d.$$

Deforming the coefficient contour to $w = -1 - x$ is justified by its $O(w^{-q-2} \log^d w)$ decay and integrability at the branch endpoint. Folding $x \leftrightarrow 1/x$ yields

$$\begin{aligned} c_{a,N}(-q-1, k) &= \frac{(-1)^q N!}{\pi k! d!} \int_1^\infty \frac{\mathcal{R}_a(x)^k P_d(\log x)}{(1+x)^{N+1}} \\ &\quad \cdot \left[x^{ad} + (-1)^{(a-1)k+d-1} x^q \right] dx. \end{aligned} \quad (11.1)$$

The signs follow from $\mathcal{R}_a(1/x) = (-1)^{a-1} x^{-a} \mathcal{R}_a(x)$ and $P_d(-\tau) = (-1)^{d-1} P_d(\tau)$. In particular the reflection line is $q = ad$, with a forced zero when $(a-1)k + d$ is even.

Theorem 11.1. *For every integer $a \geq 1$, all $k, q \geq 0$, and $d \in \{1, 2\}$:*

- *if $(a-1)k + d$ is odd, the coefficient is nonzero with sign $(-1)^q$;*

- if $(a-1)k+d$ is even, it vanishes exactly at $q = ad$, and otherwise has sign $(-1)^q \operatorname{sgn}(ad-q)$.

Proof. For $x > 1$, $\mathcal{R}_a(x) > 0$. Also $P_1(\log x) = \pi > 0$ and $P_2(\log x) = 2\pi \log x > 0$. The remaining factor in (11.1) is either a strictly positive sum or a difference of the fixed strict sign stated in the theorem. $\square$

[merge] At $a = 2$, where d is the host's deficit m , Theorem 11.1 is the host's Theorems 3.16 and 3.17 (xxb:thm:m1, xxb:thm:m2), proved there by the same pairing, and the depths one and two of its first-four-depth classification (Theorem 3.23); (11.1) is the host's branch-cut formula (its Proposition 3.30, xxb:dc:prop:integral) after $x = 1/t$, with $P_d(\tau) = \pi R_d(\tau)$ for the host's R_d ; and the reflection line $q = 2d$ is the host's all-depth symmetry holes (its Theorem 3.29). At $a = 1$, where d is the Lehmer–Comtet column, the theorem is the host's Proposition 2.15 (column 1 has no zero, column 2 only $b(5, 2) = 0$), and the reflection line $q = d$ is its symmetry zeros $b(4h + 1, 2h) = 0$ (Proposition 2.8). These are second routes. For $a \geq 3$ the theorem is new; it supplies the symmetry centre that the host's Question 8 asks to rederive.

12 The exponent $a = 1$: OEIS A293239 and the Lehmer–Comtet triangle (source 31)

[merge] This section is source 31, *The Leading Asymptotic of OEIS A293239: sparse zeros of the Lehmer–Comtet triangle*, without its integer-triangle lemma (printed as Lemma 9.1) and its closing section (printed in Sections 19 and 20). Its main statement is Theorem 3.1 at $a = 1$ and is printed as Corollary 12.1; the rest of the section is source 31's proof, a second route to that case. It was written about half an hour before source 28 and does not cite it. What it adds at $a = 1$ is the explicit picture: the identification of the missing terms with Lehmer–Comtet zeros, a complete classification of the critical points of a one-parameter integral (one real fold, $e^{T_*} = 1 + T_* + T_*^2$, $\rho_* \approx 1.29843$), the real-saddle law (12.24), and the curvature value $S''(2)$. Source 28's coordinates at $a = 1$ are $H_{1;d,k,q} = b(M, d)$ with $M = N - k = d + q + 1$, so source 31's ratio is $\rho = M/d = 1 + \eta + 1/d$. In this section h and j in $c_N(h, j)$ are the x -deficit and the logarithm exponent; elsewhere j is an x -exponent. The ratio that source 31 writes r (equal to ρ) is written ω (Section 2.2).

12.1 The count and the theorem

Let $a(N)$ be the number of distinct nonzero terms after collecting like monomials in

$$\frac{d^N}{dx^N} x^x = x^x \sum_{h,j \geq 0} c_N(h, j) x^{-h} (\log x)^j.$$

Put $\Phi(w) = (1 + w) \log(1 + w)$ and define the first-kind Lehmer–Comtet numbers by

$$b(M, d) = \frac{M!}{d!} [w^M] \Phi(w)^d, \quad 0 \leq d \leq M. \quad (12.1)$$

Here $b(0, 0) = 1$ and $b(M, 0) = 0$ for $M > 0$. These classical numbers are OEIS A008296; their relation to derivatives of x^x is recorded in the primary literature and in the inspected repository [5, 4, 1].

Corollary 12.1 (Source 31; Theorem 3.1 at $a = 1$). *The zeros of the Lehmer–Comtet triangle satisfy*

$$Z(N) := \#\{(M, d) : 1 \leq d \leq M \leq N, b(M, d) = 0\} = o(N^2). \quad (12.2)$$

Consequently the derivative-term sequence OEIS A293239 satisfies

$$a(N) = \frac{N^2}{2} + o(N^2).$$

This is an asymptotic statement; no global power-saving remainder or effective order threshold is asserted.

[merge] By (12.4) below, the corollary is equivalent to Theorem 3.1 at $a = 1$, with the same (ineffective) strength. Its proofs: source 28's, in Sections 3–9, which at $a = 1$ reduce to the endpoint paragraph of Section 7; and source 31's, which is the rest of this section.

To obtain the exact counting bridge, expand at a fixed $x > 0$:

$$\frac{(x+u)^{x+u}}{x^x} = \exp(u \log x) \exp(x\Phi(u/x)).$$

Coefficient extraction gives

$$\frac{d^N}{dx^N} x^x = x^x \sum_{M=0}^N \binom{N}{M} (\log x)^{N-M} \sum_{d=0}^M b(M, d) x^{d-M}. \quad (12.3)$$

The pair of powers $(N - M, d - M)$ determines (M, d) uniquely. Thus

$$a(N) = 1 + \frac{N(N+1)}{2} - Z(N). \quad (12.4)$$

It remains to prove (12.2). The known reflection zeros $b(4j+1, 2j) = 0$ and the sporadic zero $b(8, 5) = 0$ are fully compatible with it. We do not prove these are the only zeros.

[merge] The expansion (12.3) and the count (12.4) are the host's Theorems 2.3 and 2.7 (`xx:thm:expansion`, `xx:thm:count`), re-derived here; the host writes $b(m, r)$ with r the column and $Z(n)$ for the same zero count. The reflection zeros are the host's Proposition 2.8, and $b(8, 5) = 0$ is the exception it proves in Part 2. The host's Conjecture 2.9 asserts that these are the only zeros; the corollary does not decide it.

12.2 The integral and the complete critical-point classification

Fix $d \geq 1$ and $M > d$, and write $\rho = M/d > 1$. The change of variable $w = e^t - 1$ in (12.1) gives

$$b(M, d) = \frac{M!}{d!} \frac{1}{2\pi i} \oint \exp(dL_\rho(t)) A(t) dt, \quad (12.5)$$

$$\begin{aligned} L_\rho(t) &= \log t + t - \rho \log(e^t - 1) \\ &= \log t + (1 - \rho/2)t - \rho \log \sinh(t/2) - \rho \log 2, \end{aligned} \quad (12.6)$$

$$A(t) = \frac{e^t}{e^t - 1}.$$

The full integer-parameter integrand is single-valued; its poles other than zero lie at $2\pi i\mathbb{Z} \setminus \{0\}$. The saddle equation is

$$F(t) := (1 + 1/t)(1 - e^{-t}) = \rho. \quad (12.7)$$

12.2.1 A positive kernel

For the modulus put

$$\omega = \rho, \quad e = \frac{|\rho - 2|}{2}, \quad h = \frac{\omega}{2} - e = \min(1, \rho - 1) > 0.$$

Here e is a tilt magnitude; Euler's number is denoted e . Reflect the real coordinate when $1 < \rho < 2$, so the favored direction is left. Up to a constant phase the reflected action is

$$L_c(t) = \log t - et - \omega \log \sinh(t/2) - \omega \log 2.$$

For $t = -s - iy$, $s > 0$, $0 < y < \pi$, define

$$U(s, y) = \frac{s \coth s - y \cot y}{s^2 + y^2}, \quad V(s) = \frac{\omega}{2} - e \coth s = h - \frac{2e}{e^{2s} - 1}. \quad (12.8)$$

The classical partial-fraction expansions of $\coth$ and $\cot$ [6] give

$$U(s, y) = 2 \sum_{n \geq 1} \frac{(n\pi)^2}{(s^2 + (n\pi)^2)((n\pi)^2 - y^2)}. \quad (12.9)$$

In particular $U_s < 0$, $U_y > 0$, and $U(s, 0)$ decreases from $1/3$ to zero. If $\rho \neq 2$, then $V' > 0$ and V increases from $-\infty$ to h . There is a unique $s_0 > 0$ such that $U(s_0, 0) = V(s_0)$. For every $s > s_0$ there is a unique $y(s) \in (0, \pi)$ satisfying $U(s, y(s)) = V(s)$. It is strictly increasing and tends to π .

Introduce the auxiliary critical map

$$G(t) = \frac{\sinh t}{t} - \frac{\omega}{2}(1 + \cosh t) - e \sinh t, \quad L'_c(t) = \frac{G(t)}{\sinh t}. \quad (12.10)$$

Direct algebra gives

$$\operatorname{Im} G(-s - iy) = \sinh s \sin y [U(s, y) - V(s)].$$

Along the curve $U = V$, the Cauchy–Riemann equations imply

$$\begin{aligned} \operatorname{Re} G'(-s - iy) &= -\sinh s \sin y U_y < 0, \\ \frac{d}{ds} \operatorname{Re} G(-s - iy(s)) &= -\operatorname{Re} G'(-s - iy(s))(1 + y'(s)^2) > 0. \end{aligned} \quad (12.11)$$

The real critical value tends to $+\infty$: indeed $U = V \rightarrow h$ implies $y \rightarrow \pi$, and substitution into (12.10) gives $e^{-s}G(-s - iy(s)) \rightarrow h/2$. Thus the critical curve runs increasingly from $G(-s_0)$ to infinity, and all its roots are simple.

12.2.2 The single real fold

For $\rho > 2$ one has

$$G(-s) = \frac{\sinh s}{s} - \sinh s - \frac{\rho}{2}(1 + e^{-s}) \leq \frac{\sinh s}{s} - (1 + \cosh s) < 0,$$

since $\tanh(s/2) < s$. Hence a unique favored complex pair exists for every $\rho > 2$.

For $1 < \rho < 2$, the original favored side is positive real. On this axis

$$G(-T) = \frac{e^T + 1}{2}(F(T) - \rho), \quad F'(T) = \frac{e^{-T}(T^2 + T + 1) - 1}{T^2}. \quad (12.12)$$

The equation

$$e^{T_*} = 1 + T_* + T_*^2 \quad (12.13)$$

has exactly one positive solution. To see this, differentiate $e^T - 1 - T - T^2$ twice: its second derivative changes sign once, forcing first a decrease and then a single crossing back to positive values. The inequalities $e < 3$ and $e^2 > 7$ show $1 < T_* < 2$. Thus F increases from 1 to

$$\rho_* := F(T_*) = \frac{(T_* + 1)^2}{T_*^2 + T_* + 1} \in (1, 2)$$

and decreases to 1. Also $dG(-T)/dT = \sinh T[U(T, 0) - V(T)]$, so $G(-T)$ has its unique maximum at s_0 . Its sign at that maximum is the sign of $\rho_* - \rho$, by (12.12). Combining this fact with (12.11) proves the classification:

- if $1 < \rho < \rho_*$, there are two positive real saddles, and the smaller one satisfies $T < s_0$;
- if $\rho > \rho_*$ and $\rho \neq 2$, there is exactly one favored complex pair in $0 < |\operatorname{Im} t| < \pi$;
- at $\rho = \rho_*$ there is a real double saddle.

At $\rho = 2$, the lower simple saddle is $t = -iT$, where

$$\tan(T/2) = T, \quad \pi/2 < T < \pi. \quad (12.14)$$

The positive solution is unique because $\tan(T/2)/T$ is strictly increasing there, begins below one, and diverges. Moreover

$$L_2''(-iT) = \frac{1 - T^2}{2T^2} < 0, \quad \operatorname{Re} \frac{dt}{d\rho} \Big|_{\rho=2} = \frac{1}{2L_2''(-iT)} < 0. \quad (12.15)$$

The implicit continuation is therefore on the favored side for both signs of $\rho - 2$. Uniqueness identifies it with the preceding branches. The complex-saddle domain is the connected interval (ρ_*, ∞) .

12.3 Global selection by explicit contours

The classification alone does not identify a Cauchy contribution. We next provide contours on which the selected saddles are the only global modulus maxima.

12.3.1 The tanh coordinate

Set $z = \tanh(t/2)$, so $t = 2 \operatorname{arctanh} z$ is analytic and univalent on $|z| < 1$. Exactly,

$$A(t) dt = \frac{dz}{z(1-z)}. \quad (12.16)$$

Consequently

$$b(M, d) = \frac{M!}{d!} 2^{d-M} [z^M](\operatorname{arctanh} z)^d (1+z)^d (1-z)^{M-d-1}. \quad (12.17)$$

Every circle $|z| = R < 1$ represents the exact original Cauchy cycle. Its image is a simple contour around zero in $|\operatorname{Im} t| < \pi/2$.

On a favored quadrant write $t = -s - iy$. Since

$$|\tanh(t/2)|^2 = \frac{\cosh s - \cos y}{\cosh s + \cos y},$$

the circle is described by $\cos y / \cosh s = (1 - R^2)/(1 + R^2) > 0$. Along its arc,

$$y'(s) = -\tanh s \cot y < 0.$$

Direct differentiation, without imposing stationarity, gives

$$\frac{d}{ds} \operatorname{Re} L_c(-s - iy(s)) = \tanh s [U(s, y(s)) - V(s)]. \quad (12.18)$$

The bracket strictly decreases, because its derivative is $U_s + U_y y' - V' < 0$. The left side tends to $e > 0$ at the imaginary endpoint.

For any favored complex saddle with height below $\pi/2$, choose $R = |\tanh(t_s/2)|$. Its equation $U = V$ places it at the unique strict maximum of the favored quadrant. At that point the

second derivative in (12.18) is strictly negative. Reflection to the other half-contour lowers the real action by $2e|\operatorname{Re} t|$. The two conjugate favored saddles are therefore the only global maxima, with nonzero Gaussian curvature.

For the smaller positive real saddle $T < s_0$, choose $R = \tanh(T/2)$. Every interior point of the favored arc satisfies

$$U(s, y) > U(s, 0) > V(s), \quad s < T.$$

Thus the arc is strictly increasing up to T , which is the unique global maximum on the full contour. Its angular curvature is positive, since $F'(T) > 0$.

12.3.2 Horizontal contours for high saddles

Suppose $\pi/2 \leq y < \pi$. On the line $t = -s - iy$, put $C = \cosh s$, $S = \sinh s$, $c = \cos y$ and $D = s^2 + y^2$. For $J(s) = \operatorname{Re} L_c(-s - iy)$,

$$J'(s) \frac{C - c}{S} = B_1(s) + eB_2(s) - \frac{\omega}{2}, \quad B_1 = \frac{s(C - c)}{DS}, \quad B_2 = \frac{C - c}{S}. \quad (12.19)$$

Both B_1 and B_2 strictly decrease. Indeed

$$B_2'(s) = \frac{Cc - 1}{S^2} < 0,$$

and the periodized partial-fraction identity gives

$$\frac{1}{B_1(s)} = 2 \sum_{n \in \mathbb{Z}} \frac{s^2 + y^2}{s^2 + (y + 2\pi n)^2}. \quad (12.20)$$

Every summand with $n \neq 0$ strictly increases, because $|y + 2\pi n| > y$. The series and its differentiated series converge locally uniformly. The right side of (12.19) decreases from $+\infty$ to $-h$. It has a unique zero, the favored saddle, which is the unique favored-side maximum. Reflection strictly suppresses the other side.

Close the horizontal strip by vertical segments and let their real coordinates tend to infinity. Its only enclosed pole is zero. The estimate

$$\operatorname{Re} L_c(t) \leq \log(1 + |\operatorname{Re} t|) - h|\operatorname{Re} t| + O(1)$$

and boundedness of A on the two horizontal lines justify the limit and exponential tail control. Thus the lower and upper favored saddles are the only contributing maxima. Their horizontal Gaussian curvature has positive real part. All these bounds are uniform on compact parameter intervals away from the real fold, $\rho = 2$, and, if desired, the contour-switch points where $|\operatorname{Im} t_s| = \pi/2$.

12.4 The additive cosine law and its phase curvature

Use the original, unreflected lower saddle t_s , and write $S(\rho) = L_\rho(t_s)$. On a $\tanh$ circle define

$$z_s = \tanh(t_s/2), \quad C_s = \sinh^2(t_s) L_\rho''(t_s), \quad P_c = \frac{1}{(1 - z_s)\sqrt{C_s}}, \quad (12.21)$$

where the square root has positive real part. The curvature satisfies $\operatorname{Re} C_s > 0$. Parameterizing $z = Re^{i\theta}$ in (12.16) gives the Cauchy amplitude $1/(1 - z)$, which explains the entire factor in (12.21). On a horizontal contour the lower-saddle factor is instead

$$P_h = -i \frac{A(t_s)}{\sqrt{-L_\rho''(t_s)}}. \quad (12.22)$$

The lower boundary is traversed left to right, accounting for $-i$. Consistent local branches give the same saddle contribution when the contours can be compared.

Proposition 12.2. *On every compact interval in the complex-saddle region admitting one of the preceding contour choices,*

$$b(M, d) = \frac{2M!}{d!} \frac{e^{d \operatorname{Re} S} |P|}{\sqrt{2\pi d}} \left\{ \cos(d \operatorname{Im} S + \arg P) + O(d^{-1}) \right\}, \quad (12.23)$$

where $P = P_c$ or P_h . The error is uniform and additive, including at cosine zeros. The stationary action and the nonzero prefactor are locally analytic in ρ . For a compact interval in $1 < \rho < \rho_*$, the selected real saddle gives

$$b(M, d) = \frac{M!}{d!} \frac{e^{dL_\rho(T)}}{(1 - \tanh(T/2))\sqrt{2\pi d C_T}} \{1 + O(d^{-1})\} > 0 \quad (12.24)$$

for all sufficiently large d , where $C_T = \sinh^2(T) L''_\rho(T) > 0$.

Proof. The strict contour maxima provide a uniform gap off fixed local saddle neighborhoods. Inside a neighborhood the action is analytic with nonzero Gaussian curvature and the amplitude is analytic. Taylor expansion under the rescaling $\theta = d^{-1/2}u$ gives the leading Gaussian integral. Odd terms integrate to zero; the first remaining error is $O(d^{-1})$ relative to the single-saddle amplitude. The far arcs and horizontal tails are exponentially small. Summing the conjugate contributions gives (12.23); no division by its cosine occurs. For a real saddle there is one contribution and all factors in (12.24) are positive. $\square$

Stationarity and the affine dependence of L on ρ give

$$S'(\rho) = -\log(e^{t_s} - 1), \quad S''(\rho) = -\frac{e^{t_s}}{(e^{t_s} - 1)F'(t_s)}. \quad (12.25)$$

At $\rho = 2$, relation (12.14) implies

$$e^{-iT} = \frac{1 - iT}{1 + iT}, \quad F'(-iT) = \frac{1 - T^2}{T(T + i)}, \quad S''(2) = \frac{1}{2} + i \frac{T}{T^2 - 1}.$$

Thus $\operatorname{Im} S''(2) > 0$. The real-analytic function $\operatorname{Im} S''$ is not identically zero on the connected complex interval (ρ_*, ∞) ; its zeros are discrete. The same holds for the artificial height-switch set, since the height near $\rho = 2$ is strictly greater than $\pi/2$.

12.5 Sparse integer points and exhaustion of the triangle

We include the elementary counting argument so that the density conclusion does not depend on a presumed equidistribution theorem.

[merge] Source 31 states and proves here, as its Lemma 5.1, the integer-triangle estimate printed once as Lemma 9.1; its proof, with the two details only source 31 spells out, is given there.

On a compact ratio interval where $|\operatorname{Im} S''|$ is bounded below, define for each fixed d

$$f_d(M) = \frac{d \operatorname{Im} S(M/d) + \arg P(M/d)}{\pi} - \frac{1}{2}.$$

An exact zero in (12.23) satisfies $\operatorname{dist}(f_d(M), \mathbb{Z}) \leq C/d$. Also

$$f_d''(M) = \frac{\operatorname{Im} S''(M/d)}{\pi d} + O(d^{-2}),$$

so Lemma 9.1 applies. The number of zero cells with such ratios is $O(d^{2/3})$ per depth and $O(N^{5/3})$ after summing over $d \leq N$. Compact intervals in the real-saddle region have no zeros for large d .

Fix a compact subinterval of $(1, \infty)$. First remove arbitrarily small neighborhoods of the real fold ρ_* and of $\rho = 2$. The remaining complex portions are compactly contained in (ρ_*, ∞) , so they contain only finitely many curvature or contour-switch zeros. Cover these by additional intervals, with the total removed length less than ε . The lattice count in the removed intervals is $O(\varepsilon N^2) + O_\varepsilon(N)$. Cover the remaining compact set by finitely many intervals to which the preceding estimates apply. Taking $N \rightarrow \infty$ and then $\varepsilon \downarrow 0$ shows that the zero cells have density zero on every compact ratio interval.

Finally restrict the full candidate triangle to

$$\varepsilon N \leq d \leq M \leq N, \quad M/d \geq 1 + \varepsilon.$$

The ratios now lie in $[1 + \varepsilon, 1/\varepsilon]$. The excluded candidates have area $O(\varepsilon)$ after scaling by N , and their boundary contributes only $O(N)$. The compact-ratio zero count is $o(N^2)$. Taking $N \rightarrow \infty$ first and $\varepsilon \downarrow 0$ afterward proves (12.2). Equation (12.4) completes the proof of Corollary 12.1.

13 The exponent $a = 2$: OEIS A290268 (source 34)

[merge] This section is source 34, *The Leading Asymptotic of OEIS A290268*, with its input file `tanh_sections.tex` printed in place as Sections 13.4–13.6, without its integer-triangle lemma (printed as Lemma 9.1) and with its closing section printed in Sections 19 and 20. Its main statement is Theorem 3.1 at $a = 2$ and is printed as Corollary 13.1; the rest is source 34's proof, a second route. Source 34 was the first of the three proofs to be written, about an hour before source 28, and it does not cite source 28. It adds at $a = 2$: the monotone critical curve (Lemma 13.2), a direct proof that every complex saddle with $\eta > 2$ is high, real-saddle positivity on the whole strict real region (Theorem 13.9), and a rational certificate for the curvature seed. Its coordinates are source 28's at $a = 2$: $H_{d,k,q} = H_{2;d,k,q}$, and its action (13.8) is (4.2) at $a = 2$, because $\sinh t = 2 \sinh(t/2) \cosh(t/2)$. Renamed here (Section 2.2): the ratio $r = \eta + 2$ is ω ; the host's $r = 2d + q + 1$, which source 34 writes R , is $M - k$; $A(s)$, $B(s)$ are $B_1(s)$, $B_2(s)$; the circle parameter and radius are α and R ; the curvature $C(\kappa, \eta)$ is $\text{Im } D^2 S$; and $K(T)$, $F(T)$ are $\mathcal{K}(T)$, $\mathcal{F}(T)$. Source 34's Laurent polynomial $P_N(x, \log x)$ is not the host's canonical polynomial $P_n(z, L)$.

13.1 The result and the coefficient model

Define the Laurent polynomial in the formal symbols x and $\log x$ by

$$\frac{d^N}{dx^N} x^{x^2} = x^{x^2} P_N(x, \log x), \quad P_N = \sum_{j \in \mathbb{Z}, k \geq 0} c_N(j, k) x^j (\log x)^k,$$

and let $a(N)$ count its nonzero coefficients.

Corollary 13.1 (Source 34; Theorem 3.1 at $a = 2$). *For the derivative-term sequence OEIS A290268,*

$$\lim_{N \rightarrow \infty} \frac{a(N)}{N^2} = \frac{1}{2}.$$

More precisely, the outside-bulk tail candidate set described below has only $o(N^2)$ exact zero coefficients. No numerical error rate or order threshold is asserted, and individual extra cancellations are not classified.

[merge] The corollary is Theorem 3.1 at $a = 2$, with the same (ineffective) strength. Its proofs: source 28's, in Sections 3–9, and source 34's, which is the rest of this section. It answers the host's Question 6. It does not supersede the host's explicit lower bounds, which have leading term $3N^2/8$ and hold for every N .

The source [1] gives, for $d \geq 1$ and $k, q \geq 0$,

$$M = k + 2d + q + 1, \quad (13.1)$$

$$H_{d,k,q} = \frac{M!}{d!} [w^M] (2+w)^k (1+w)^{2d} \log^d(1+w), \quad (13.2)$$

$$N = 2k + 2d + q + 1, \quad j = -q - 1, \quad c_N(j, k) = \binom{N}{k} H_{d,k,q}. \quad (13.3)$$

The bridge is injective in (d, k) at fixed N . The positive bulk already proved in the source consists of the nonnegative x -power columns and the positive $j = -1$ boundary. Its exact size is

$$\beta(N) = \begin{cases} (3N^2 + 10N + 8)/8, & N \text{ even,} \\ (3N^2 + 12N + 1)/8, & N \text{ odd.} \end{cases} \quad (13.4)$$

Cells with $q \geq 1$ lie outside this bulk. Depth zero contributes no additional negative-column cells: its coefficient reduces to $[w^{k+q+1}](2+w)^k = 0$. The proved reflection holes are $q = 2d$, $k + d$ even; their existence is consistent with the density theorem.

An upper bound is immediate and requires no noncancellation claim. The recurrence

$$P_{N+1} = \frac{d}{dx} P_N + x(2 \log x + 1) P_N, \quad P_0 = 1, \quad (13.5)$$

preserves the constraints

$$\ell = (N + j)/2 \in \mathbb{Z}, \quad 0 \leq k \leq \ell \leq N.$$

Indeed multiplication by x raises ℓ by one and raises k by either zero or one; differentiation leaves ℓ fixed and leaves k fixed or lowers it by one. Thus

$$a(N) \leq \frac{(N+1)(N+2)}{2} = \frac{N^2}{2} + O(N). \quad (13.6)$$

The work below proves the matching lower bound.

[merge] The model (13.2)–(13.3) is the host's depth-coordinate coefficient (its Section 3.14); the bulk is the host's nonvanishing region (Theorem 3.7), and $\beta(N)$ is the host's $\Lambda(N)$ (Theorem 3.1) and $\beta(N)$ (equation (3.85)); the reflection holes are the host's Theorem 3.29. The upper bound (13.6) is weaker than the host's Theorem 3.14, $a(N) \leq U(N) = N^2/2 + N + 1$, which source 34 does not cite; it suffices here. The recurrence (13.5), read on coefficients, is the recurrence by which the repository's Lean library defines `LeanProofs.A290268.coeff` (its lemma `coeff_succ`; Section 1.3).

13.2 An exact contour and the favored critical points

Put

$$\kappa = k/d, \quad \eta = q/d, \quad \delta = \eta - 2, \quad \omega = \eta + 2.$$

The substitution $w = e^t - 1$ in (13.2) gives exactly

$$H_{d,k,q} = \frac{M!}{d!} \frac{1}{2\pi i} \oint \frac{\exp(dL(t))}{4 \sinh^2(t/2)} dt, \quad (13.7)$$

where

$$L(t) = \log t - \frac{\delta}{2}t + \kappa \log \cosh(t/2) - (\kappa + \omega) \log \sinh(t/2) - \omega \log 2. \quad (13.8)$$

For example, the unsimplified integrand is $2^{-(M-k)-1}t^d e^{-(q-2d)t/2} \cosh^k(t/2)/\sinh^{M+1}(t/2)$. The extra powers in the denominator include the Cauchy differential. The original integer powers make the integrand single-valued despite local logarithm choices; its other poles are at $2\pi i\mathbb{Z} \setminus \{0\}$.

We first work off the line $\eta = 2$. Write e for Euler's number, and let

$$e = |\eta - 2|/2 > 0, \quad h = \omega/2 - e = \min(2, \eta) > 0.$$

The left half-plane is favored in modulus when $\eta > 2$, and the right half-plane when $0 < \eta < 2$. Reflection of the real coordinate reduces both cases to

$$L_c(t) = \log t - et + \kappa \log \cosh(t/2) - (\kappa + \omega) \log \sinh(t/2) - \omega \log 2. \quad (13.9)$$

For $0 < \eta < 2$, the reflected parameters are $h = \eta$, $\omega = \eta + 2$, $e = (2 - \eta)/2$. Reflection preserves real action values and sends a lower-left critical point of the favored model to the original lower-right point.

On the lower favored quadrant write $t = -s - iy$, $s > 0$, $0 < y < \pi$. The classical partial fractions [6] give

$$U(s, y) = \frac{s \coth s - y \cot y}{s^2 + y^2} = 2 \sum_{n \geq 1} \frac{(n\pi)^2}{(s^2 + (n\pi)^2)((n\pi)^2 - y^2)}. \quad (13.10)$$

The series and its derivatives converge locally uniformly. Hence $U_s < 0$, $U_y > 0$, $U(s, 0)$ decreases from $1/3$ to zero, and $U(s, y) \rightarrow \infty$ as $y \uparrow \pi$.

The critical equation is $F_{\omega, e}(t) = \kappa$, with

$$F_{\omega, e}(t) = \frac{\sinh t}{t} - \frac{\omega}{2}(1 + \cosh t) - e \sinh t.$$

Its imaginary part is

$$\operatorname{Im} F_{\omega, e}(-s - iy) = \sinh s \sin y \{U(s, y) - V(s)\}, \quad V(s) = h - \frac{2e}{e^{2s} - 1}. \quad (13.11)$$

The function V increases from $-\infty$ to $h > 0$. There is one $s_0 > 0$ with $U(s_0, 0) = V(s_0)$, and for $s > s_0$ one curve $y(s) \in (0, \pi)$ solving $U(s, y) = V(s)$. It increases strictly from zero to π .

Lemma 13.2 (Monotone favored critical curve). *Along this curve, the real critical value $\kappa(s) = \operatorname{Re} F_{\omega, e}(-s - iy(s))$ increases strictly to infinity. Every interior critical point on it is simple.*

Proof. At a zero of the imaginary part,

$$\operatorname{Re} F'_{\omega, e}(-s - iy) = -\sinh s \sin y U_y < 0.$$

The Cauchy–Riemann equations along the real-value curve imply

$$\kappa'(s) = -\operatorname{Re} F'_{\omega, e}(-s - iy)(1 + y'(s)^2) > 0.$$

As $s \rightarrow \infty$, the height tends to π and the real critical value has a positive term of leading size $he^s/2$; the term $\sinh t/t$ is smaller by $O(1/s)$. Thus $\kappa(s) \rightarrow \infty$. The strict derivative also makes $F'_{\omega, e}$ nonzero. Since $L'_c = (F_{\omega, e} - \kappa)/\sinh t$ in the open strip, the saddle is simple. $\square$

For $\eta > 2$, every $\kappa \geq 0$ has a favored complex saddle of height greater than $\pi/2$. Here $h = 2$. With $C = \cosh s$, $c = \cos y$, $D = s^2 + y^2$, eliminating e gives

$$\frac{\kappa}{C+c} = e^s \left\{ \frac{s + y(e^{-s} - c)/\sin y}{D} - 2 \right\}. \quad (13.12)$$

At $y = \pi/2$, the numerator after subtracting $2D$ decreases with s , because its derivative is $1 - (y/\sin y)e^{-s} - 4s < 1 - e^{-s} - s < 0$. Its value at zero is $y - 2y^2 < 0$. Thus the height- $\pi/2$ crossing has negative κ , and monotonicity proves the assertion.

13.3 Horizontal contours for the high complex saddles

Lemma 13.3 (Global horizontal selection). *Every favored complex saddle of height $\pi/2 \leq y < \pi$ is the unique global modulus maximum on its lower horizontal line. The upper line has the conjugate maximum, and both horizontal Hessians are nondegenerate.*

Proof. In the reflected favored model, put $J_y(s) = \operatorname{Re} L_c(-s - iy)$, $C = \cosh s$, $S = \sinh s$, and $c = \cos y \leq 0$. Differentiation yields

$$J'_y(s) \frac{C-c}{S} = B_1(s) + eB_2(s) - \frac{\omega}{2} - \frac{\kappa c}{C+c},$$

where

$$B_1(s) = \frac{s(C-c)}{(s^2 + y^2)S}, \quad B_2(s) = \frac{C-c}{S}.$$

The partial-fraction identity

$$\frac{1}{B_1(s)} = 2 \sum_{n \in \mathbb{Z}} \frac{s^2 + y^2}{s^2 + (y + 2\pi n)^2}$$

makes B_1 strictly decreasing: the $n = 0$ term is constant and every other term increases. Also $B'_2(s) = (C \cos y - 1)/S^2 < 0$, while the last term is nonincreasing. The entire right side therefore decreases strictly from $+\infty$ to $e - \omega/2 = -h < 0$. Its unique zero is the classified saddle, and $J''_y < 0$ there.

At the reflected point on the other half-line, the real action is smaller by $2es$. Thus the selected point is the unique maximum on the whole line. In original coordinates this gives $\operatorname{Re}(-L''(t_s)) > 0$. $\square$

The small contour (13.7) deforms to the boundary of $|\operatorname{Im} t| \leq y$, with vertical sides tending to infinity. No other pole is enclosed. Uniformly on compact parameter sets, the real action on the ends is at most $\log(1 + |\operatorname{Re} t|) - h|\operatorname{Re} t| + C$, so the vertical integrals vanish and the tails are negligible. The unique maxima and their Hessians give a uniform gap away from the two local saddle arcs.

For the original lower favored saddle define

$$P_h = \frac{-i}{4 \sinh^2(t_s/2) \sqrt{-L''(t_s)}}, \quad \Theta_h = d \operatorname{Im} L(t_s) + \arg P_h. \quad (13.13)$$

The square root has positive real part. The resulting compact-uniform asymptotic is

$$H_{d,k,q} = \frac{2M!}{d! \sqrt{2\pi d}} e^{d \operatorname{Re} L(t_s)} |P_h| \{\cos \Theta_h + O(d^{-1})\}. \quad (13.14)$$

The lower horizontal integral is traversed left to right, the upper right to left. They are conjugates, so division by $2\pi i$ extracts the lower imaginary part and produces the factor $-i$. Expanding the analytic exponent through its cubic term and the prefactor through its linear term, odd Gaussian terms cancel; the remaining normalized moments are $O(d^{-1})$. The error in (13.14) is additive, including at cosine zeros.

13.4 Tanh circles select the low-height saddles

The remaining contour is particularly simple after one more change of coordinate. It is a circle in the variable $z = \tanh(t/2)$, although its image in the t -plane is bent.

Lemma 13.4 (Exact tanh-coordinate integral). *Let $\widehat{L}(z) = L(2 \operatorname{arctanh} z)$. For every $0 < R < 1$,*

$$H = \frac{M!}{d!} \frac{1}{2\pi i} \int_{|z|=R} \frac{e^{\widehat{dL}(z)}}{2z^2} dz. \quad (13.15)$$

Equivalently,

$$H = \frac{M!}{d!} 2^{-d-q-1} [z^M](1+z)^{2d}(1-z)^q \operatorname{arctanh}(z)^d. \quad (13.16)$$

The full integrand in (13.15) is single-valued at the integer parameters. The image of the circle is a bounded simple positively oriented contour around zero in $|\operatorname{Im} t| < \pi/2$.

Proof. The inverse map $t = 2 \operatorname{arctanh} z = \log((1+z)/(1-z))$ is analytic and univalent on the unit disk: the fraction in the logarithm maps that disk onto the right half-plane. In particular its image lies in $|\operatorname{Im} t| < \pi/2$. Directly,

$$\frac{dt}{4 \sinh^2(t/2)} = \frac{dz}{2z^2}. \quad (13.17)$$

Furthermore $\omega d = 2d + q$ and $\delta d = q - 2d$, so

$$e^{\widehat{dL}(z)} = \frac{(2 \operatorname{arctanh} z)^d (1+z)^{2d} (1-z)^q}{2^{2d+q} z^{k+2d+q}}.$$

Substitution in the exact t -integral proves both identities, including the coefficient index $M = k + 2d + q + 1$. Inside the disk the only singularity of the full integrand is its pole at zero, so each circle represents the original residue contour. $\square$

[merge] The coefficient identity (13.16) is source 28's (3.6) at $a = 2$, and it is the host's transformed coefficient formula (its Proposition 3.6, `xxb:prop:mobius`), which source 34 does not cite, written in depth coordinates ($k_{\text{host}} = k + d$, $j_{\text{host}} = k$, $m = d$). The batch-72A intake checked the host's formula and (13.16) against the exact coefficients for every $n \leq 16$ (for the latter, on all 168 cells with $q \geq 0$, $d \geq 1$), with no mismatch; that check is not shipped.

For the modulus use the reflected favored model

$$L_c(t) = \log t - et + \kappa \log \cosh(t/2) - (\kappa + \omega) \log \sinh(t/2) - \omega \log 2, \\ e = |\delta|/2, \quad h = \omega/2 - e > 0.$$

Here the favored half-plane is the left half-plane. Reflection changes only phase conventions, not the real-action comparison; the final normalizations below are always in the original coordinate.

Lemma 13.5 (The bent-arc derivative). *On the lower favored arc of $|\tanh(t/2)| = R < 1$, put $t = -s - iy$. There is an $\alpha \in (0, 1)$ such that*

$$\frac{\cos y}{\cosh s} = \alpha = \frac{1 - R^2}{1 + R^2}, \quad 0 \leq s \leq s_{\text{end}} := \operatorname{arcosh}(1/\alpha), \quad y'(s) = -\tanh s \cot y. \quad (13.18)$$

For $J(s) = \operatorname{Re} L_c(-s - iy(s))$, one has the exact identity

$$J'(s) = \tanh s [U(s, y(s)) - V(s)], \quad V(s) = \frac{\omega}{2} - e \coth s = h - \frac{2e}{\exp(2s) - 1}. \quad (13.19)$$

If $e > 0$, the bracket is strictly decreasing on the open arc. Thus its modulus profile either increases throughout or has exactly one critical point, a strict global maximum.

Proof. Writing $C = \cosh s$, $S = \sinh s$, and $c = \cos y$, the identity

$$|\tanh((-s - iy)/2)|^2 = \frac{C - c}{C + c}$$

gives (13.18). For $u(s, y) = \operatorname{Re} L_c(-s - iy)$, differentiation gives

$$\begin{aligned} u_s &= \frac{s}{s^2 + y^2} + e + \frac{\kappa}{2} \frac{S}{C + c} - \frac{\kappa + \omega}{2} \frac{S}{C - c}, \\ u_y &= \frac{y}{s^2 + y^2} - \frac{\kappa \sin y}{2(C + c)} - \frac{\kappa + \omega}{2} \frac{\sin y}{C - c}. \end{aligned}$$

In $u_s + y' u_y$, all κ terms cancel. What remains is

$$\tanh s \left[\frac{s \coth s - y \cot y}{s^2 + y^2} - \frac{\omega}{2} + e \coth s \right],$$

which proves (13.19). The kernel signs in (13.10) give $U_s < 0$ and $U_y > 0$; also $V'(s) = e \operatorname{csch}^2 s > 0$. Consequently

$$\frac{d}{ds}[U(s, y(s)) - V(s)] = U_s + U_y y' - V' < 0. \quad (13.20)$$

At the initial endpoint $J'(s) \rightarrow e > 0$. The assertion follows. $\square$

Theorem 13.6 (Global selection below height $\pi/2$). *Suppose $e > 0$, $h > 0$, and a favored off-axis saddle has height strictly between 0 and $\pi/2$. Its circle in the z -coordinate selects it and its conjugate as the only global maxima of the leading modulus. The angular Gaussian curvature has strictly positive real part at each.*

Proof. In reflected coordinates write the lower saddle as $t_s = -s_s - iy_s$ and choose $R = |\tanh(t_s/2)| < 1$. Its imaginary critical equation is $U(s_s, y_s) = V(s_s)$, by (13.11). Lemma 13.5 therefore makes it the unique maximum on the lower favored arc, with

$$J''(s_s) = \tanh(s_s)[U_s + U_y y' - V'] < 0.$$

The arc parameter is regular at this interior point, so the same strict sign holds for the real part of the angular second derivative. The upper favored arc is its conjugate. Both axis endpoints are strictly below the two interior maxima. At every point in the other open half-contour, reflection to the favored side increases the real action by $2e|\operatorname{Re} t|$. No further global maximum is possible. $\square$

Proposition 13.7 (Low-height two-arc normalization). *In the original, unreflected coordinate let t_s be the lower favored saddle from Theorem 13.6, and put*

$$z_s = \tanh(t_s/2), \quad C_s = z_s^2 \widehat{L}''(z_s) = \sinh(t_s)^2 L''(t_s), \quad (13.21)$$

$$P_c = \frac{1}{2z_s \sqrt{C_s}}, \quad \Theta_c = d \operatorname{Im} L(t_s) + \arg P_c. \quad (13.22)$$

The square root has positive real part. Uniformly on every compact family of such saddles,

$$H = A_c \{\cos \Theta_c + O(d^{-1})\}, \quad A_c = \frac{2(M!/d!)e^{d \operatorname{Re} L(t_s)} |P_c|}{\sqrt{2\pi d}} > 0. \quad (13.23)$$

This is an additive error estimate, including at the cosine zeros. The stationary action and the nonzero prefactor are locally analytic functions of the parameters.

Proof. Parameterize the z -circle by its positive angle. The residual amplitude in (13.15) is then $1/(2z)$. At a saddle, $\widehat{L}' = 0$ and $z dt/dz = \sinh t$, proving (13.21). The angular second derivative is $-C_s$, and Theorem 13.6 gives $\operatorname{Re} C_s > 0$.

On a fixed compact parameter family, all relevant derivatives are bounded, the local negative quadratic bound is uniform, and the strict modulus gap off the two saddle neighborhoods is uniform. Taylor expansion on symmetric angular neighborhoods of radius $d^{-2/5}$, followed by Gaussian integration, has relative error $O(d^{-1})$ for each single saddle contribution: the terms of order $d^{-1/2}$ are odd and integrate to zero. The other portions of those neighborhoods are $O(\exp(-cd^{1/5}))$ on the leading scale, and the arcs outside them have a fixed exponential gap. The lower contribution is

$$\frac{M! e^{dL(t_s)} P_c}{d! \sqrt{2\pi d}} \{1 + O(d^{-1})\}.$$

The upper contribution is its conjugate. Their sum proves (13.23) without division by the leading cosine. Local analyticity follows from the simple critical-point equation and from the nonvanishing factors in P_c . $\square$

Where the horizontal and bent contours are both admissible, these are the same local Gaussian contributions. With consistent lower-saddle branches, $\sqrt{C_s} = i \sinh(t_s) \sqrt{-L''(t_s)}$, and hence

$$P_c = \frac{-i}{4 \sinh^2(t_s/2) \sqrt{-L''(t_s)}}.$$

This is exactly the horizontal prefactor. Local branch choices suffice for all phase-counting arguments.

13.5 The entire strict real-saddle region

For $0 < \eta < 2$ the favored direction in the original coordinate is the positive real direction. Set

$$f_\eta(T) = \frac{\sinh T}{T} - e^{-T} - 1 - \frac{\eta}{2}(1 + e^T). \quad (13.24)$$

The real saddle equation is $f_\eta(T) = \kappa$. Here $e = (2 - \eta)/2$, $\omega = \eta + 2$, and $h = \eta$ in the reflected modulus model.

Lemma 13.8 (A unique real turning point). *There is exactly one $s_0 = s_0(\eta) > 0$ with $U(s_0, 0) = V(s_0)$. The function f_η increases strictly on $(0, s_0)$ and decreases strictly on (s_0, ∞) . Its maximum*

$$\kappa_0(\eta) = f_\eta(s_0(\eta)) \quad (13.25)$$

is analytic for $0 < \eta < 2$.

Proof. The function $U(T, 0)$ decreases from $1/3$ to zero, whereas $V(T)$ increases from $-\infty$ to $\eta > 0$. Direct differentiation of (13.24) gives

$$f'_\eta(T) = \sinh T [U(T, 0) - V(T)]. \quad (13.26)$$

This proves the unique turning point and the derivative signs. The strict derivative of $U(T, 0) - V(T)$ also proves analytic dependence by the implicit function theorem. $\square$

Theorem 13.9 (Unrestricted smaller-real-saddle positivity). *If $0 < \eta < 2$ and $0 \leq \kappa < \kappa_0(\eta)$, the smaller positive solution T of $f_\eta(T) = \kappa$ is selected by its $\tanh$ circle as the unique global modulus maximum. Put*

$$C_T = \sinh(T)^2 L''(T) = \sinh(T) f'_\eta(T) > 0.$$

Uniformly on every compact subset of this strict parameter region,

$$H = \frac{(M!/d!)e^{dL(T)}}{2 \tanh(T/2) \sqrt{2\pi d C_T}} \{1 + O(d^{-1})\} > 0 \quad (13.27)$$

for sufficiently large d . No upper bound on T is required.

Proof. Since $f_\eta(0) = -1 - \eta < 0$ and $f_\eta(T) \rightarrow -\infty$, the assumed strict inequality gives a unique smaller root $T \in (0, s_0)$. Choose $R = \tanh(T/2)$. On the favored bent arc ending at T , every interior point has $s < T$ and

$$U(s, y(s)) > U(s, 0) > V(s).$$

Thus the profile in Lemma 13.5 is strictly increasing up to its real endpoint. Reflection strictly suppresses the unfavored half-contour, making T the unique global maximum. The curvature identity follows from $L' = (f_\eta - \kappa)/\sinh T$ at the real saddle. It is positive by (13.26). The one-arc Gaussian calculation is the same as in Proposition 13.7, with positive amplitude $1/(2R)$ and with no factor of two. This proves (13.27); all of its leading factors are positive. $\square$

13.6 Coverage and connected phase regions

For $0 < \eta < 2$, the favored off-axis critical curve starts at $(s_0, 0)$ and its real critical value increases strictly from $\kappa_0(\eta)$ to infinity, by Lemma 13.2. Consequently every $\kappa \geq 0$ strictly below the turning value is covered by Theorem 13.9; every κ strictly above it has one favored conjugate pair. Theorem 13.6 selects that pair below height $\pi/2$, and the horizontal theorem selects it at larger height.

At height exactly $\pi/2$, the horizontal proof also remains strict: the term proportional to $\cos y$ vanishes while the other derivative terms remain strictly negative. Alternatively this contour-switch curve can be omitted from a density exhaustion, as it has area zero. For $\eta > 2$ the earlier horizontal result already selects every favored saddle. The lines $\eta = 2$, $\eta = 0$, and $\kappa = 0$, together with the real turning curve, also have area zero.

13.7 Compact density away from the analytic exceptional sets

The two complex-saddle regions are

$$\begin{aligned} \mathcal{C}_- &= \{0 < \eta < 2, \kappa > 0, \kappa > \kappa_0(\eta)\}, \\ \mathcal{C}_+ &= \{\eta > 2, \kappa > 0\}. \end{aligned}$$

Both are connected. For $\mathcal{C}_-$, raise κ above the maximum of the continuous κ_0 on the compact interval between any two chosen η values, move across that interval, and lower it again. The second region is immediate. On each, the favored saddle and its stationary action $S(\kappa, \eta) = L(t_s(\kappa, \eta))$ are analytic.

The directional phase curvature relevant to a fixed derivative order is

$$\operatorname{Im} D^2 S(\kappa, \eta) = (\partial_\kappa - 2\partial_\eta)^2 \operatorname{Im} S, \quad D = \partial_\kappa - 2\partial_\eta. \quad (13.28)$$

It is not identically zero in either connected region. To check this without numerical evidence, put

$$\mathcal{K}(T) = \frac{\sin T}{T} - 2 \cos T - 2, \quad \mathcal{F}(T) = (2T^2 - 1) \sin T + T \cos T.$$

At $T = 14/5$, alternating Taylor bounds give $\mathcal{K}(T) > 1/250$ and $\mathcal{F}(T) > 2$. Thus $\eta = 2$, $\kappa = \mathcal{K}(T)$ has a nondegenerate saddle $t_s = -iT$ with $L''(t_s) < 0$. Its continuation enters the favored side as $\eta \rightarrow 2^\pm$, since $\operatorname{Re}(\partial_\eta t_s) = 1/(2L''(t_s)) < 0$.

The action is affine in κ, η , and $L_{t\kappa} - 2L_{t\eta} = 1 + \coth t$. Stationary differentiation gives

$$(\partial_\kappa - 2\partial_\eta)^2 S = -\frac{(1 + \coth t_s)^2}{L''(t_s)}.$$

At the displayed base point its imaginary part is $2T^2 \cos T / \mathcal{F}(T) < 0$. Therefore (13.28) is a nonzero real-analytic function on each complex region.

[merge] Source 34 states and proves here, as its Lemma 7.1, the integer-triangle estimate printed once as Lemma 9.1, whose proof records its choice $Mc^3/2 < 1/4$ of the block constant.

On any compact complex patch with $\text{Im } D^2 S$ of constant sign bounded away from zero, the selected contour gives

$$H = \mathcal{A}\{\cos(d \text{Im } S + \arg P) + O(d^{-1})\}, \quad \mathcal{A} > 0,$$

with analytic nonzero prefactor P . At fixed N, d let

$$\eta = \frac{N - 1 - 2d - 2k}{d}, \quad f_{d,N}(X) = \frac{1}{\pi} \left[d \text{Im } S \left(\frac{X}{d}, \frac{N - 1 - 2d - 2X}{d} \right) + \arg P \right] - \frac{1}{2}.$$

Its second derivative is bounded between fixed positive multiples of $1/d$ in one sign. An exact zero forces $\text{dist}(f_{d,N}(k), \mathbb{Z}) = O(d^{-1})$, by the additive cosine law. Lemma 9.1 bounds zeros by $O(d^{2/3})$ per row. Over a fixed compact parameter patch, $d \asymp N$ and there are $O(N)$ rows, yielding $O(N^{5/3})$ zeros.

The remaining exceptional sets have area zero: the line $\eta = 2$, the real turning curve $\kappa = \kappa_0(\eta)$, the contour-switch curve $\text{Im } t_s = -\pi/2$, and the zero set of (13.28). The first two are explicit analytic curves. The latter two are zero sets of nonzero real-analytic functions on connected complex regions; nontriviality of the height condition follows from the same regular limit. A nonzero real-analytic function has an area-zero zero set: at each zero a finite-order directional derivative is nonzero, and Rolle's theorem followed by Fubini's theorem proves the assertion locally. A countable cover completes the proof.

Now fix a compact rectangle Ω inside $\kappa > 0, \eta > 0$. Cover the exceptional set in its scaled lattice image by finitely many open boxes of total area less than ϵ . If necessary first cover the turning and switch boundaries, then the remaining compact analytic zero sets. Outside those boxes there is a finite cover by complex patches of nonzero curvature and real-saddle patches. The latter have no zero coefficients once N is large, uniformly on each compact patch. The former have $O_\epsilon(N^{5/3})$ zeros. The boxes contain at most $\epsilon(N - 1)^2 + O_\epsilon(N)$ candidates. Taking $N \rightarrow \infty$ first and then $\epsilon \downarrow 0$ proves that every compact Ω has only $o(N^2)$ exact cancellations.

13.8 Exhausting the full tail and completing the count

The ratio-to-lattice map is

$$x = \frac{d}{N - 1} = \frac{1}{2 + 2\kappa + \eta}, \quad y = \frac{k}{N - 1} = \frac{\kappa}{2 + 2\kappa + \eta}, \quad \left| \frac{\partial(x, y)}{\partial(\kappa, \eta)} \right| = (2 + 2\kappa + \eta)^{-3}. \quad (13.29)$$

The image of a fixed compact rectangle has piecewise smooth boundary. Its candidate count is its area times $(N - 1)^2$, with $O(N)$ boundary error. Exhaust the parameter cone by $\Omega_m = [1/m, m]^2$. The areas increase to

$$\int_0^\infty \int_0^\infty (2 + 2\kappa + \eta)^{-3} d\eta d\kappa = \frac{1}{8}. \quad (13.30)$$

All analytic constants are fixed before N tends to infinity; there is no global uniform power-saving error claim.

The complete outside-bulk candidate list has $d \geq 1$, $k \geq 0$, $q \geq 1$ and $N = 2k + 2d + q + 1$. Let $m_N = \lfloor (N - 2)/2 \rfloor$. Its exact size is

$$\sum_{s=1}^{m_N} s = \frac{m_N(m_N + 1)}{2} = \frac{N^2}{8} + O(N). \quad (13.31)$$

For each fixed m , the compact patch Ω_m has only $o(N^2)$ zeros. The number of candidates outside it, divided by N^2 , has limsup at most $1/8 - I_m$, where I_m is its area under (13.29). Letting $m \rightarrow \infty$ proves that the entire tail candidate set has only $o(N^2)$ zeros. Boundary rows such as $k = 0$ are absorbed by this exhaustion and its exact total count.

The nonzero tail therefore has $N^2/8 + o(N^2)$ cells. These give distinct monomials by (13.3), all with $j \leq -2$, so they are disjoint from (13.4). Hence

$$a(N) \geq \beta(N) + \frac{N^2}{8} + o(N^2) = \frac{N^2}{2} + o(N^2).$$

Together with the recurrence upper bound (13.6), this proves Corollary 13.1.

Part II

The half exponent

14 Derivative supports of $x^{\sqrt{x}}$ (source 26)

[merge] This section is source 26, *Derivative Supports of $x^{\sqrt{x}}$: a leading-density theorem with two infinite extra-cancellation families*, without its integer-triangle lemma (printed as Lemma 9.1) and with its closing section printed in Sections 19 and 20. Its exponent $a = 1/2$ is not an integer, so Theorem 14.1 is not a case of Theorem 3.1; the method is the same family, adapted. Source 26 was written about twenty minutes after source 28 and alludes to it (“the integer-exponent horizontal proof”) without citing it. Its ratio $\eta = 2\nu + \kappa - 1$ is written $\hat{\eta}$ here, because it is not source 28’s $\eta = q/d$, and its favoured real point $z = -r$ is $z = -\sigma$, because r is the defect in Section 14.6. Its $\nu = (N - k)/d = M/d$ plays the role of source 28’s radial exponent ν in (6.1), and its direction $D = \partial_\kappa - \partial_\nu$ the role of source 28’s D . The two families of Theorem 14.3 are, in Part III’s language, defects two and three at $a = 1/2$.

14.1 The theorem and an exact reciprocal coefficient

For $x > 0$, define coefficients by

$$\frac{d^N}{dx^N} x^{\sqrt{x}} = x^{\sqrt{x}} \sum_{\ell, k} p_{N, \ell, k} x^{\ell/2 - N} (\log x)^k. \quad (14.1)$$

For $N \geq 1$, the potential cells are $1 \leq \ell \leq N$, $0 \leq k \leq \ell$, with distinct monomials. Let $A(N)$ be the number of nonzero coefficients.

Theorem 14.1. *The number of zero candidates in (14.1) is $o(N^2)$. Consequently*

$$A(N) = \frac{N^2}{2} + o(N^2).$$

No global power-saving remainder or effective order threshold is asserted. The exact cancellation families in Section 14.6 show that reflection is not the only zero mechanism.

Put $d = \ell - k \geq 0$. Taylor expansion gives

$$p_{N,\ell,k} = \frac{N!}{k!d!} [w^N] (\sqrt{1+w} - 1)^k (1+w)^{d/2} \log^d(1+w). \quad (14.2)$$

Use the reciprocal coordinate

$$\sqrt{1+w} = \frac{1+z}{1-z}, \quad w = \frac{4z}{(1-z)^2}, \quad dw = \frac{4(1+z)}{(1-z)^3} dz.$$

Then $\sqrt{1+w} - 1 = 2z/(1-z)$ and $\log(1+w) = 4 \operatorname{arctanh} z$. The exact result is

$$p_{N,\ell,k} = \frac{N!}{k!d!} 2^k 4^{d-N} [z^{N-k}] (\operatorname{arctanh} z)^d (1+z)^{d+1} (1-z)^{2N-k-d-1}. \quad (14.3)$$

The map $4z/(1-z)^2$ is univalent from the unit disk onto the plane cut along $(-\infty, -1]$, with the displayed inverse and branches. Every circle $|z| = R < 1$ is therefore an exact coefficient cycle.

For the interior $d > 0$, introduce

$$\nu = \frac{N-k}{d} > 1, \quad \kappa = \frac{k}{d} > 0, \quad \hat{\eta} = 2\nu + \kappa - 1 > 1. \quad (14.4)$$

Writing

$$\Psi(z) = \log \operatorname{arctanh} z + \log(1+z) + \hat{\eta} \log(1-z) - \nu \log z, \quad (14.5)$$

the residual differential in (14.3) is

$$\frac{1+z}{z(1-z)} dz. \quad (14.6)$$

All outside constants in (14.3) are positive. The original integer powers make the full integrand single-valued despite local logarithm choices.

14.2 Circle-only global saddle selection

Set $t = 2 \operatorname{arctanh} z$. Up to the real constant $-\log 2$, the action is

$$L(t) = \log t - \frac{\hat{\eta}-1}{2} t + (\nu - \hat{\eta} - 1) \log \cosh(t/2) - \nu \log \sinh(t/2). \quad (14.7)$$

The effective cosh exponent is $\nu - \hat{\eta} - 1 = -\nu - \kappa < 0$. Thus no integer-exponent horizontal monotonicity theorem is being assumed.

On $|z| = R < 1$, the favored side is left. Write $t = -s - iy$ in its lower quadrant. The circle satisfies

$$\frac{\cos y}{\cosh s} = \frac{1-R^2}{1+R^2} > 0, \quad y'(s) = -\tanh s \cot y < 0.$$

Define

$$U(s, y) = \frac{s \coth s - y \cot y}{s^2 + y^2}, \quad V(s) = \frac{\hat{\eta}+1}{2} - \frac{\hat{\eta}-1}{2} \coth s.$$

The partial-fraction identity [6]

$$U(s, y) = 2 \sum_{n \geq 1} \frac{(n\pi)^2}{(s^2 + (n\pi)^2)((n\pi)^2 - y^2)}$$

implies $U_s < 0$, $U_y > 0$ and $V' > 0$. The radial power in (14.5) has constant modulus on the circle. Direct differentiation gives

$$\frac{d}{ds} \operatorname{Re} L(-s - iy(s)) = \tanh s [U(s, y(s)) - V(s)]. \quad (14.8)$$

The bracket strictly decreases because $U_s + U_y y' - V' < 0$. The derivative is positive near the imaginary endpoint. Thus each favored quadrant has one maximum, either a strict interior maximum or its negative real endpoint. Reflection lowers the real action by $(\hat{\eta} - 1)|\operatorname{Re} t|$, so the opposite side is strictly suppressed. Every selected complex saddle has nonzero angular curvature.

14.2.1 The positive terminal derivative

Let $\mathcal{M}(\xi)$ be the maximum of $\operatorname{Re} \Psi$ on $|z| = e^\xi$, $\xi < 0$. The modulus in (14.5) is subharmonic on annuli; comparison with a harmonic affine function of $\log |z|$ proves convexity of $\mathcal{M}$. Its initial slope is $1 - \nu < 0$.

As $R \uparrow 1$, the maxima remain at bounded s , because the favored-end action is at most $\log(1 + s) - s + O(1)$. The limiting circle is the line $y = \pi/2$. On that line $U(s, \pi/2) - V(s)$ strictly decreases from $+\infty$ to -1 , giving a unique limiting horizontal maximum. The circle maxima converge to it.

For $u(s, y) = \operatorname{Re} L(-s - iy)$, differentiation of (14.7) at $y = \pi/2$ gives

$$u_y = \frac{\pi}{2(s^2 + \pi^2/4)} - \frac{2(\nu - \hat{\eta} - 1) + \hat{\eta} + 1}{2 \cosh s} = \frac{\pi}{2(s^2 + \pi^2/4)} + \frac{\kappa}{2 \cosh s} > 0. \quad (14.9)$$

Since $dt/d\xi = \sinh t = -i \cosh s$ at this line, the envelope has strictly positive terminal derivative. Convexity now forces a stationary minimum strictly inside $R < 1$.

This minimum is unique. At a complex angular maximum, harmonicity gives strictly positive radial envelope curvature. A flat zero-slope interval could therefore contain only real stationary maxima, which would make the analytic action derivative vanish on an interval and hence identically, contradicting its pole at zero. Conjugate angular maxima have the same radial derivative, so the envelope is differentiable even when the maximum changes between the real endpoint and a conjugate pair.

We have obtained a selected real saddle or conjugate pair for every interior physical parameter using exact circles only. Except at degenerate real saddles, the selected saddle is simple, the global gap is strict, and the geometry persists uniformly on compact parameter patches.

14.3 A real threshold and one connected complex region

Write the favored real point as $z = -\sigma$, $0 < \sigma < 1$. Its logarithmic radial derivative is

$$\begin{aligned} & \frac{\sigma}{(1 - \sigma^2) \operatorname{arctanh} \sigma} - \frac{\sigma}{1 - \sigma} + \frac{\hat{\eta} \sigma}{1 + \sigma} - \nu \\ &= \frac{1 - \sigma}{1 + \sigma} [E(\sigma; \kappa) - \nu], \end{aligned} \quad (14.10)$$

where

$$E(\sigma; \kappa) = f(\sigma) + \kappa g(\sigma), \quad f(\sigma) = \frac{\sigma}{(1 - \sigma)^2} \left(\frac{1}{\operatorname{arctanh} \sigma} - 2 \right), \quad g(\sigma) = \frac{\sigma}{1 - \sigma}. \quad (14.11)$$

One has $f(\sigma) < 1$, since $\sigma / \operatorname{arctanh} \sigma < 1$. Also

$$f(\sigma) = 1 - \frac{4}{3} \sigma^2 + O(\sigma^3), \quad E(\sigma; \kappa) \longrightarrow -\infty \quad (\sigma \uparrow 1).$$

For each $\kappa > 0$, the linear term $\kappa \sigma$ makes $E > 1$ near zero. Hence

$$M(\kappa) = \max_{0 < \sigma < 1} E(\sigma; \kappa) > 1 \quad (14.12)$$

exists, is continuous and nondecreasing, and tends to one as $\kappa \downarrow 0$. The dominant negative $(1 - \sigma)^{-2}$ term confines maximizers away from $\sigma = 1$ on bounded κ intervals. On compact subintervals, $f < 1$ and $\kappa g \rightarrow 0$ uniformly; together with the limit $f(0) = 1$, this proves the last assertion and continuity at zero.

When $1 < \nu < M(\kappa)$, a first simple real crossing has $E_\sigma > 0$. Equation (14.10) gives positive radial curvature and hence a local angular maximum. The uniqueness proved in (14.8) makes it the selected global real maximum. For $\nu > M(\kappa)$ there is no favored real stationary point, so the selected saddle is complex. This complex region is the connected epigraph above M .

The degenerate real locus has area zero. The equations $E = \nu$ and $E_\sigma = 0$ solve affinely for (κ, ν) because $g'(\sigma) = (1 - \sigma)^{-2} > 0$; their solutions form an analytic curve image. On compact parameter rectangles away from $\nu = 1$, $\kappa = 0$, real stationary points stay away from both $\sigma = 0$ and $\sigma = 1$, so the exceptional image is compact there. Nongeneric tangencies do not produce an additional open complex component, by the stability of a selected complex saddle and the density of parameters with simple first real crossings.

A selected negative-real saddle contributes a single nonzero term. Its sign is $(-1)^{N-k-d}$, coming from $(\operatorname{arctanh} z)^d z^{-(N-k)}$; positivity is not required for the density argument.

14.4 A compact-uniform cosine law and its curvature seed

Let z_s be the selected lower complex saddle and put

$$S = \Psi(z_s), \quad C_s = z_s^2 \Psi''(z_s), \quad P = \frac{1 + z_s}{(1 - z_s)\sqrt{C_s}}. \quad (14.13)$$

The square root is fixed by the positively oriented Gaussian circle, with $\operatorname{Re} C_s > 0$. The ratio $(1 + z_s)/(1 - z_s)$ is the full amplitude obtained from (14.6).

Proposition 14.2. *On every compact simple complex-saddle patch,*

$$p_{N,\ell,k} = \frac{N!}{k!d!} 2^k 4^{d-N} \frac{2e^{d\operatorname{Re} S} |P|}{\sqrt{2\pi d}} \cdot \left\{ \cos(d\operatorname{Im} S + \arg P) + O(d^{-1}) \right\}. \quad (14.14)$$

The error is additive and compact-uniform. On a compact simple real-saddle patch, the analogous one-saddle expansion has a nonzero leading term.

Proof. The selected circle has a strict global modulus gap and nonzero local Gaussian curvature. Taylor expansion under the $d^{-1/2}$ rescaling and cancellation of odd Gaussian terms give a relative $O(d^{-1})$ error for each individual saddle. The complementary arcs are exponentially smaller. Summing the conjugate contributions gives (14.14); the error is never divided by the cosine. $\square$

At fixed N, d , increasing k follows $D = \partial_\kappa - \partial_\nu$. Affine parameter dependence and stationarity imply

$$DS = \log \frac{z_s}{1 - z_s}, \quad D^2 S = -\frac{1}{z_s^2 (1 - z_s)^2 \Psi''(z_s)}. \quad (14.15)$$

We give an explicit boundary expansion to prove this curvature is not identically zero.

Set $\kappa = 0$, $\nu = 1 + \varepsilon$, $\hat{\eta} = 1 + 2\varepsilon$. The saddle equation is exactly

$$F(z, \varepsilon) = \frac{z}{\operatorname{arctanh} z} - 1 - \varepsilon - 2\varepsilon z - (1 + \varepsilon)z^2 = 0, \quad \Psi_z = \frac{F}{z(1 - z^2)}. \quad (14.16)$$

Let $\varepsilon = h^2$. Since

$$\frac{F(hv, h^2)}{h^2} = -1 - \frac{4}{3}v^2 - 2hv + O(h^2),$$

the analytic implicit-function theorem at the lower root gives

$$z = -i\frac{\sqrt{3}}{2}h - \frac{3}{4}h^2 + O(h^3). \quad (14.17)$$

Its negative real part identifies the favored side. At $\kappa = 0$, the inequality $f < 1$ excludes a real saddle for every $\nu > 1$, so uniqueness identifies this local branch with the selected complex saddle.

At that saddle, (14.15) simplifies to

$$D^2S = -\frac{1+z}{z(1-z)F_z}.$$

If $v_0 = -i\sqrt{3}/2$, then $F_z = -(8/3)hv_0 + O(h^3)$: the h^2 term cancels because the next coefficient in (14.17) is $-3/4$. Substitution gives

$$D^2S = -\frac{1}{2\varepsilon} + i\frac{\sqrt{3}}{4\sqrt{\varepsilon}} + O(1). \quad (14.18)$$

Its imaginary part is positive for sufficiently small fixed $\varepsilon > 0$. It persists at small positive κ because $M(\kappa) \rightarrow 1$. Connectedness of the complex region and real analyticity now show that $\text{Im } D^2S$ has a zero set of area zero.

14.5 Density in the full candidate triangle

We recall the elementary integer-triangle estimate in the form needed here.

[merge] Source 26 states and proves here, as its Lemma 5.1, the integer-triangle estimate printed once as Lemma 9.1.

On a compact patch avoiding the phase-curvature zero set, an exact zero in (14.14) lies within $O(1/d)$ of a half-integer normalized phase. Its second derivative in integer k , with d, N fixed, is

$$\frac{\text{Im } D^2S}{\pi d} + O(d^{-2}).$$

Lemma 9.1 gives $O(d^{2/3})$ zeros per row and $O(N^{5/3})$ on the compact patch. Simple real-saddle patches contain no zeros for large d .

First remove neighborhoods of the boundaries $\nu = 1$ and $\kappa = 0$, and restrict to compact parameter rectangles. Remove a small-area neighborhood of the compact real-degenerate locus before taking finite small-area covers of the remaining analytic exceptional set. The complement has a finite simple-saddle cover. Taking $N \rightarrow \infty$ first, then the removed area to zero, proves zero density on every compact physical rectangle.

The ratio-to-lattice map is

$$\left(\frac{d}{N}, \frac{k}{N}\right) = \left(\frac{1}{\nu + \kappa}, \frac{\kappa}{\nu + \kappa}\right),$$

with Jacobian $(\nu + \kappa)^{-3}$. Its full area is

$$\int_1^\infty \int_0^\infty \frac{d\kappa d\nu}{(\nu + \kappa)^3} = \frac{1}{2}.$$

This is exactly the interior area of $1 \leq \ell \leq N$, $0 \leq k \leq \ell$, with $\ell = k + d$. The boundaries $d = 0$, $k = 0$, and $\ell = N$ contain only $O(N)$ cells. Compact exhaustion therefore gives $o(N^2)$ zero candidates in the full triangle. Since its exact size is $N(N+3)/2$, Theorem 14.1 follows.

14.6 Two infinite nonreflection zero families

The defect $r = N - \ell$ counts the loss in the number of exponent factors. It is different from the depth $d = \ell - k$. Let $g(x) = \sqrt{x} \log x$ and $L = \log x$. The required low derivatives are

$$\begin{aligned} g' &= x^{-1/2}(1 + L/2), & g'' &= x^{-3/2}(-L/4), \\ g''' &= x^{-5/2}(3L/8 - 1/4), & g'''' &= x^{-7/2}(1 - 15L/16). \end{aligned}$$

At defect one, the exponential Bell expansion contains one g'' and $N - 2$ copies of g' . Thus, for $N \geq 2$, the unique zero on $\ell = N - 1$ is $k = 0$. This is the reciprocal-reflection family.

Theorem 14.3. *For every $N \geq 3$ and $0 \leq k \leq N - 2$,*

$$p_{N,N-2,k} = \frac{2^{-k}N(N-1)}{96} \binom{N-2}{k} (3k^2 + 13k - 4N + 8). \quad (14.19)$$

Its zeros are exactly

$$N = \frac{3k^2 + 13k + 8}{4}, \quad k \geq 1, \quad k \equiv 0, 1 \pmod{4}. \quad (14.20)$$

For every $N \geq 4$ and $0 \leq k \leq N - 3$,

$$p_{N,N-3,k} = -\frac{2^{-k}N(N-1)(N-2)}{384} \binom{N-3}{k} (k+4)(k^2 + 9k - 4N + 12). \quad (14.21)$$

Its zeros are exactly

$$N = \frac{k^2 + 9k + 12}{4}, \quad k \geq 3, \quad k \equiv 0, 3 \pmod{4}. \quad (14.22)$$

Both families are infinite and lie outside the reflection line.

Proof. For defect two the block-size patterns are 3 and $2 + 2$. With $u = L/2$, the corresponding logarithmic polynomial is

$$\binom{N}{3} \frac{(1+u)^{N-4}}{16} (3(N+1)u^2 + 8u - 4).$$

Coefficient extraction and cancellation of the binomial factors give (14.19). The same identity holds directly at $N = 3$, using g''' , so no division at a vanishing binomial denominator is used.

For defect three the patterns are 4, $3 + 2$, and $2 + 2 + 2$. Their multiplicities are $\binom{N}{4}$, $10\binom{N}{5}$, and $15\binom{N}{6}$, respectively. The logarithmic polynomial becomes

$$\binom{N}{4} \frac{(1+u)^{N-6}}{16} (16 + 2(2N-7)u - 4(2N+3)u^2 - (N+1)(N+2)u^3).$$

Extracting coefficients gives (14.21); $N = 4, 5$ follow directly from the available block patterns.

The zero equations reduce to the quadratics displayed in (14.20) and (14.22). Reduction modulo four gives their congruence classes. At defect two, $k = 0$ formally gives $N = 2$, outside the $N \geq 3$ domain; every listed $k \geq 1$ is admissible. At defect three, $k = 0$ gives $N = 3$, outside $N \geq 4$; the listed $k \geq 3$ are all admissible. The reflection condition is $N = (k + 2d)/2 + 1$, which at defect r reads $r + k/2 = 1$. It has no admissible point at defects two or three. $\square$

The first members of (14.20) are $(N, k) = (6, 1), (27, 4), (37, 5), (76, 8)$; those of (14.22) are $(12, 3), (16, 4), (31, 7), (37, 8)$. These sparse families approach the boundary corner $(\kappa, \nu) = (0, 1)$, so their existence is consistent with the compact zero-density proof.

[merge] Theorem 14.3 is also a case of source 20r's general residuals, by a different derivation. In the notation of Section 15, at $a = 1/2$ the reflection form is $K = k/2 + r - 1$ and $c = -1$, and the exact residuals (16.12) give

$$R_{1/2,2,N} = \frac{3k^2 + 13k + 8 - 4N}{96}, \quad R_{1/2,3,N} = -\frac{(k+4)(k^2 + 9k + 12 - 4N)}{384},$$

which by the bridge (15.3) are (14.19) and (14.21). Source 20r's reflection line (16.11) at $a = 1/2$ is $k = -2(r - 1)$, physical only on defect one, where it is the reciprocal-reflection family $k = 0$ above; 26's condition $r + k/2 = 1$ is $K = 0$. Source 20r's Corollary 16.3 on infinite nonreflection families concerns the open interval $1/2 < a < 1$; at the endpoint $a = 1/2$ only the positive branch of each parabola is physical, and source 20's endpoint remark after Theorem 17.2 counts these two families through order X as $\sqrt{X}/3 + O(1)$ and $\sqrt{X} + O(1)$. None of the three sources cites the others. The verification of Section 20 finds, for $N \leq 150$, sixteen nonreflection zeros, all in these two families (six and ten).

Part III

Fixed defects for non-integer exponents

[merge] Part III is sources 20r, 20 and 16, written in that order within about forty minutes. All three slice the coefficients by a fixed factor defect $r = N - \ell$ as N grows, the other natural slicing besides fixed depth. At $a = 1$ a fixed defect is a fixed Lehmer–Comtet offset, the subject of the host's Conjecture 2.9, which all three exclude; at an integer $a \geq 2$ every cell with $N \geq 2r$ lies in the positive bulk of Proposition 3.2 (at $a = 2$, the host's Theorem 3.7), so a fixed defect is eventually nonzero there for an elementary reason. The fixed-defect theorems therefore have content for non-integer a . A rational exponent is written $a = \mathfrak{p}/\mathfrak{q}$ in lowest positive terms (Section 2.2).

15 The centered coefficient identity (sources 20r, 20, 16)

[merge] Sources 20r, 20 and 16 each open with this setup in nearly the same words. It is printed once, in source 20r's words, with the sentences only source 20 or only source 16 contains. Source 16 writes the residual as $R_r(a; N, k)$, a polynomial in (a, N, k) , and the reflection form as K_a ; in this section, including source 16's Lemma 15.1, they are $R_{a,r,N}(k)$ and K , and Section 18 keeps source 16's notation. Ten equation labels of sources 20 and 16 name equations of this section and were merged into source 20r's labels (the README lists them).

For fixed $a > 0$, write

$$\frac{d^N}{dx^N} x^{x^a} = x^{x^a} \sum_{\ell, k} p_{N, \ell, k}(a) x^{a\ell - N} (\log x)^k. \quad (15.1)$$

The factor defect is $r = N - \ell$. Throughout a physical cell means

$$N \geq r + 1, \quad 0 \leq k \leq N - r, \quad N, k \in \mathbb{Z}. \quad (15.2)$$

Distinct pairs (ℓ, k) represent distinct monomials because $a > 0$.

Set

$$A_a(z) = \frac{(1+z)^a \log(1+z)}{z}, \quad B_a(z) = \frac{(1+z)^a - 1}{az}.$$

Both power series have constant term one. Taylor expansion of $x^a \log x$ and the partial Bell-polynomial formula give

$$p_{N,N-r,k}(a) = a^k \frac{N!}{(N-r)!} \binom{N-r}{k} R_{a,r,N}(k), \quad (15.3)$$

$$R_{a,r,N}(k) = [z^r] A_a(z)^{N-r-k} B_a(z)^k. \quad (15.4)$$

The prefactor in (15.3) is positive throughout (15.2). At each fixed r , the residual is a polynomial in N, k with rational coefficients when a is rational. (source 16) The coefficients of A_a and B_a are polynomials in a ; the apparent singularity of B_a at zero is removable coefficient by coefficient. Thus $R_{a,r,N}(k) \in \mathbb{Q}[a, N, k]$.

Write

$$K = ak - (2a - 1)N + 2ar - 1, \quad c = (a - 1)(2a + 1). \quad (15.5)$$

[merge] Source 20r defines K in its Theorem 1.1 and c in its Section 3; source 20 defines both together, as here, and source 16 writes K_a for K . On an odd defect, $K = 0$ is the *reflection line*.

The substitution $t = \log(1 + z)$ in the coefficient integral, including its differential, gives

$$R_{a,r,N}(k) = [t^r] \exp(-Kt/2) \left(\frac{\sinh(at/2)}{at/2} \right)^k \left(\frac{t/2}{\sinh(t/2)} \right)^{N+1}. \quad (15.6)$$

Indeed,

$$A_a = e^{(a-1/2)t} \frac{t/2}{\sinh(t/2)}, \quad B_a = e^{(a-1)t/2} \frac{\sinh(at/2)}{a \sinh(t/2)},$$

and the Cauchy differential supplies $e^{(1-r)t/2} (t/(2 \sinh(t/2)))^{r+1}$. The exponential factors combine to $e^{-Kt/2}$, and the remaining factors combine as displayed.

Both hyperbolic ratios in (15.6) are even functions. Thus the residual vanishes at $K = 0$ whenever r is odd. After expressing k linearly in N, K , this is an exact polynomial factor. Equivalently, the reciprocal substitution $z \mapsto -z/(1 + z)$ multiplies A_a by $(1 + z)^{1-2a}$ and B_a by $(1 + z)^{1-a}$, while the Cauchy differential contributes $(-1)^r (1 + z)^{r-1}$. The combined power of $1 + z$ is exactly K .

15.1 Degrees of the residual (source 16)

Lemma 15.1. *As a polynomial in (N, k) at fixed a , $R_{a,r,N}(k)$ has total degree r , with leading homogeneous part*

$$\frac{\Lambda_a(N, k)^r}{r!}, \quad \Lambda_a(N, k) = (a - 1/2)N - ak/2. \quad (15.7)$$

At every physical cell, $R_{a,r,N}(k)$ is a nonzero polynomial in a of exact degree r .

Proof. The first coefficients of $\log A_a$ and $\log B_a$ are respectively $a - 1/2$ and $(a - 1)/2$. In

$$[z^r] \exp((N - r - k) \log A_a + k \log B_a),$$

parameter degree r can come only from r linear-in- z factors. The constant shift $-r$ lowers parameter degree. This proves (15.7); the line is nonzero for every a .

Moreover $[z^j] A_a$ has degree j in a with leading coefficient $1/j!$, and $[z^j] B_a$ has degree j with leading coefficient $1/(j + 1)!$. Put $d = N - r - k \geq 0$. The coefficient of a^r in (15.4) is exactly

$$[w^r] e^{dw} \left(\frac{e^w - 1}{w} \right)^k > 0. \quad (15.8)$$

Both series have nonnegative coefficients. Since $d + k = N - r \geq 1$, at least one factor has strictly positive coefficients in every degree. This proves strict positivity and exact degree. $\square$

[merge] In this proof $d = N - r - k = \ell - k$ is the depth, as everywhere in this report, and e is Euler's number. Source 16's leading line is written Λ_a here; source 28's $L_a(t)$ is an action.

16 Rational exponents $a \neq 1$ (source 20r)

[merge] This section is source 20r, *Reflection and Finite Exceptions: fixed defects in derivatives of x^x for positive rational $a \neq 1$* , which arrived only inside archive 20 as its companion. Its coefficient setup is Section 15; its closing section is printed in Sections 19 and 20.

Theorem 16.1. *Fix $a \in \mathbb{Q}_{>0} \setminus \{1\}$ and an integer $r \geq 4$. With K as in (15.5), only finitely many physical cells have $p_{N,N-r,k}(a) = 0$, after excluding $K = 0$ when r is odd. No exclusion is needed for even r . In particular each fixed defect is eventually free of nonreflection cancellations. Every odd defect has the exact reflection factor K .*

Here “eventually” allows a threshold depending on both a and r . The proof provides no numerical threshold or complete list of the finite exceptions. It does not establish a support-density theorem when the defect grows with N .

[merge] Source 20r states the theorem in its Section 1 with the definition (15.5) of K inside it. Source 20's Theorem 17.1 (defects $r \geq 4D + 2$, so $r \geq 6$ at $D = 1$) is weaker for rational a , and source 20 says so; source 16's Theorem 18.1 covers irrational a .

16.1 The weighted edge and defects at least six

Put

$$k = \frac{K + (2a - 1)N - 2ar + 1}{a},$$

with $c = (a - 1)(2a + 1) \neq 0$ as in (15.5). Assign weight two to N and weight one to K . In the exponent of (15.6), the only terms whose parameter weight equals their t degree are

$$-Kt/2 + cNt^2/24.$$

This follows from $\log(\sinh(at/2)/(at/2)) = a^2t^2/24 + O(t^4)$ and $\log((t/2)/\sinh(t/2)) = -t^2/24 + O(t^4)$. Hence the top weighted part of the centered residual is

$$[t^r] \exp(-Kt/2 + cNt^2/24).$$

For $r = 2m$ use the whole centered residual as $F_r(N, K)$; for $r = 2m + 1$ divide it by its exact factor K . Its top weighted part is respectively

$$E_{2m}(N, K) = \frac{m!}{(2m)!} \left(\frac{cN}{6}\right)^m L_m^{(-1/2)} \left(-\frac{3K^2}{2cN}\right), \quad (16.1)$$

$$E_{2m+1}(N, K) = -\frac{m!}{2(2m+1)!} \left(\frac{cN}{6}\right)^m L_m^{(1/2)} \left(-\frac{3K^2}{2cN}\right). \quad (16.2)$$

These identities follow from the standard Hermite–Laguerre relations [7]. Both expressions are polynomials, of weighted degree $2m$ and degree m in N , with nonzero constant leading N coefficient.

Schur's theorem states that $L_m^{(-1/2)}$ and $L_m^{(1/2)}$ are irreducible over $\mathbb{Q}$ for every $m \geq 1$; a primary research account explicitly records the theorem and its original references [8, pp. 1–2]. Rational reciprocal rescaling shows that $E_r(v, 1)$ is irreducible in $\mathbb{Q}[v]$. After extending to the purely transcendental field $\mathbb{Q}(K)$ and replacing v by N/K^2 , it remains irreducible. The constant

leading N coefficient makes it primitive over $\mathbb{Q}[K]$, so Gauss's lemma gives irreducibility in $\mathbb{Q}[N, K]$.

The top weighted parts of any two polynomial factors of F_r would multiply to E_r . Since every nonconstant polynomial has positive weighted degree, irreducibility of E_r forces one such factor to be constant. Therefore F_r is irreducible over $\mathbb{Q}$.

The Laguerre roots are positive and simple. The roots of $E_r(v, 1)$ are thus distinct and nonzero, whether c is positive or negative. Write them as $\alpha_1, \dots, \alpha_m$. With $t = 1/K$, $N = v/t^2$, the expression

$$t^{2m} F_r(v/t^2, 1/t)$$

is analytic at $t = 0$ and specializes to $E_r(v, 1)$. The implicit-function theorem gives m distinct branches

$$v = v_i(t), \quad v_i(0) = \alpha_i, \quad K = t^{-1}, \quad N = t^{-2} v_i(t). \quad (16.3)$$

They are distinct points on the smooth projective normalization, and K has a simple pole at each. They exhaust the points above $K = \infty$ because the degree in N is m with constant leading coefficient.

To finish for $r \geq 6$, first suppose F_r is not absolutely irreducible. Its absolute components are Galois conjugate. A rational point lies on every conjugate of any component containing it, so it belongs to a finite intersection of distinct components. Thus there are only finitely many rational affine points in this case.

Otherwise take the smooth projective normalization C . Its affine singular points form a finite set, and nonsingular rational points lift uniquely to $C(\mathbb{Q})$. Write $a = \mathfrak{p}/\mathfrak{q}$ in lowest positive terms. At an integer cell,

$$\mathfrak{q}K = \mathfrak{p}k - (2\mathfrak{p} - \mathfrak{q})N + 2\mathfrak{p}r - \mathfrak{q} \in \mathbb{Z}.$$

By (16.3), the rational function $\mathfrak{q}K$ on C has $m \geq 3$ distinct geometric poles. The three-pole form of Siegel's theorem [9, Theorem 1.1] makes the set of rational points with $\mathfrak{q}K$ integral finite, regardless of genus. The linear change $(N, k) \mapsto (N, K)$ is injective. This proves Theorem 16.1 for every $r \geq 6$.

16.2 Parameterized quartics at defects four and five

Expansion of (15.6) through degree five uses

$$U = a^2 k - N - 1 = aK + cN + C_r, \quad C_r = -2a^2 r + a - 1, \quad V = N + 1 - a^4 k.$$

Introduce the rational functions

$$b = -\frac{2a^3 + a^2 + a + 1}{2a + 1}, \quad e = \frac{a(a + 1)}{2a + 1}, \quad f_r = -\frac{2a(a + 1)((r + 1)a - 1)}{2a + 1}. \quad (16.4)$$

Direct substitution gives $V = bU + eK + f_r$. Coefficient extraction is finite algebra:

$$5760R_{a,4,N} = 15K^4 + 30K^2U + 5U^2 + 2V, \quad (16.5)$$

$$11520R_{a,5,N} = -K(3K^4 + 10K^2U + 5U^2 + 2V). \quad (16.6)$$

Set $Y_4 = 5U + 15K^2 + b$ and $Y_5 = 5U + 5K^2 + b$. Away from $K = 0$ in the odd case, the cancellation equations become

$$Y_4^2 = 150K^4 + 30bK^2 - 10eK + b^2 - 10f_4, \quad (16.7)$$

$$Y_5^2 = 10K^4 + 10bK^2 - 10eK + b^2 - 10f_5. \quad (16.8)$$

Since $c \neq 0$, each correspondence is injective: from K, Y_r , recover U and then $N = (U - aK - C_r)/c$, followed by k from (15.5).

Lemma 16.2. *Both quartics in (16.7)–(16.8) are squarefree for every positive rational a .*

Proof. Let $Q_r(a, K)$ be the right-hand quartic multiplied by $(2a + 1)^2$, so $Q_r \in \mathbb{Z}[a, K]$. Exact expansion gives

$$\text{disc}_K Q_4 = 27000000(2a + 1)^6 D_4(a), \quad (16.9)$$

$$\text{disc}_K Q_5 = 8000(2a + 1)^6 D_5(a), \quad (16.10)$$

where D_4, D_5 are degree-18 integer polynomials. The definitions (16.4)–(16.10) specify them exactly; their full coefficient lists and an exact discriminant verifier accompany this note.

Here is a short certificate that neither has a rational root. The leading coefficient of D_4 is 512, congruent to 2 modulo 3, and

$$D_4(a) \equiv 2 \pmod{(3, a^3 - a)}.$$

The leading coefficient of D_5 is 4608, congruent to 10 modulo 11. Its remainder modulo 11 and $a^{11} - a$ is

$$\begin{aligned} & -2a^{10} - a^9 + 3a^8 + 3a^7 + 5a^6 + 5a^5 \\ & -4a^4 + 4a^3 + 2a^2 + 4a - 5. \end{aligned}$$

At $a = 0, 1, \dots, 10$ its successive values modulo 11 are

$$6, 3, 8, 8, 8, 8, 3, 2, 9, 7, 6,$$

all nonzero. Suppose a reduced rational number u/v were a root. If the selected prime divides v , the homogenized identity contradicts the nonzero leading coefficient modulo that prime. Otherwise u/v reduces to a field element, contradicting the displayed root-free values. Thus both D_r have no rational root. Finally $2a + 1 \neq 0$ for $a > 0$, so neither discriminant vanishes. $\square$

[merge] The coefficient lists of D_4 and D_5 and the exact verifier that “accompany this note” are shipped with this report as `data/20r-rational-certificates.json` and `code/20r-rational-verify.py`, with the independent replay `code/20r-rational-replay_independent.py` and its output (Section 20).

A squarefree degree-four equation $Y^2 = Q(K)$ has a genus-one smooth projective completion: the double cover of the K -line has four simple branch points. Siegel’s positive-genus theorem [10, Theorem 4], applied to the rational function $\mathfrak{q}K$, therefore makes the rational points with $\mathfrak{q}K \in \mathbb{Z}$ finite. The injective correspondence proves Theorem 16.1 for $r = 4, 5$. This completes its proof in all cases.

The modular certificates rule out exceptional *rational parameters*; they are not searches over parameter values. They do not exclude singular quartics at every irrational algebraic parameter.

16.3 Reflection lines and the first three defects

For $a = \mathfrak{p}/\mathfrak{q}$ in lowest positive terms, solving $K = 0$ over integers gives exactly

$$N = 1 + \mathfrak{p}h, \quad k = (2\mathfrak{p} - \mathfrak{q})h - 2(r - 1), \quad h \in \mathbb{Z}, \quad (16.11)$$

subject to (15.2). Indeed, reduction of $K = 0$ modulo $\mathfrak{p}$ gives $N \equiv 1 \pmod{\mathfrak{p}}$.

If $1/2 < a < 1$, then $0 < 2\mathfrak{p} - \mathfrak{q} < \mathfrak{p}$. On every fixed odd defect, all sufficiently large positive h in (16.11) are physical. These reflection cells have cumulative count $X/\mathfrak{p} + O_{a,r}(1)$. At $a = 1/2$, the line is $k = -2(r - 1)$, so only $r = 1$ supplies a physical infinite family. For $a < 1/2$ or $a > 1$, the line eventually lies outside the physical range.

The first three exact residuals are

$$R_{a,1,N} = -K/2, \quad R_{a,2,N} = (3K^2 + U)/24, \quad R_{a,3,N} = -K(K^2 + U)/48, \quad (16.12)$$

where $U = aK + cN - 2a^2r + a - 1$ uses the corresponding defect. In particular, the nonreflection zero equations at defects two and three are

$$N = -\frac{3K^2 + aK - 4a^2 + a - 1}{c}, \quad (16.13)$$

$$N = -\frac{K^2 + aK - 6a^2 + a - 1}{c}. \quad (16.14)$$

[merge] At $a = 1/2$ these residuals are source 26's exact diagonal formulas (Theorem 14.3; the computation is recorded after it). Source 26 derives them by a different route, from the exponential Bell blocks of $g(x) = \sqrt{x} \log x$.

Corollary 16.3. *For every rational $1/2 < a < 1$, defects two and three each have infinitely many physical nonreflection cancellations. On every fixed defect $r \geq 4$, all sufficiently late cancellations are precisely the physical reflection cells when r is odd, and there are no cancellations when r is even.*

Proof. Each parabola contains the structural rational point $(N, K) = (r, r - 1)$ with $k = 0$, outside the physical domain. Set $K = r - 1 + Mh$. Because the formulas for N and recovered k are rational polynomials in h with integer constant terms, a sufficiently divisible positive integer M makes both integer for every integer h . When $1/2 < a < 1$, $c < 0$, so $N \rightarrow +\infty$ as $|h| \rightarrow \infty$, while

$$\frac{k}{N} \rightarrow 2 - \frac{1}{a} \in (0, 1).$$

Consequently the cells are physical and $K \neq 0$ for all sufficiently large $|h|$. The last assertion follows from Theorem 16.1 and the exact odd factor. $\square$

[merge] Source 20's Theorem 17.2 counts the families of the first assertion exactly, up to $O(1)$. They are not reflection zeros, so the host's observation that every cancellation it proves is a parity zero (Section 1.3) does not extend to x^{x^a} for these a . In the proof, M is a positive integer and h an integer parameter, as in (16.11).

There is also an elementary description outside that interval. Uniformly for $0 \leq k \leq N - r$ at fixed a, r ,

$$R_{a,r,N}(k) = \frac{[N(a - 1/2) - ak/2]^r}{r!} + O_{a,r}(N^{r-1}). \quad (16.15)$$

This is the highest ordinary-degree part of (15.4); the fixed shift r affects only lower degree. For $a > 1$, the bracket is at least $(a - 1)N/2$, so every fixed defect is eventually positive. For $0 < a < 1/2$, it is at most $-(1/2 - a)N$, so the eventual sign is $(-1)^r$. These statements concern each fixed defect, not the whole derivative support.

[merge] The bracket in (16.15) is source 16's leading line $\Lambda_a(N, k)$ of Lemma 15.1.

The exception $a = 1$ is structural. Then $c = 0$, $K = k - N + 2r - 1$, and (15.4) becomes

$$[z^r] \left(\frac{(1 + z) \log(1 + z)}{z} \right)^{r-1-K},$$

independent of N after centering. Its possible integer roots produce entire coefficient lines, so the preceding finite-exception proof does not apply.

[merge] At $a = 1$ a fixed defect is a fixed Lehmer–Comtet offset. The batch-72A intake's exact scan (Section 19) shows such a line: the host's sporadic zero $b(8, 5) = 0$ recurs at every k , at the cells $(N, \ell, k) = (k + 8, k + 5, k)$.

17 Algebraic exponents: a degree bound and exact frequencies (source 20)

[merge] This section is source 20, *Algebraic Degrees and Exact Cancellation Frequencies: two fixed-defect extensions for derivatives of x^a* . Its Section 1 is the identity of Section 15; its closing section is printed in Sections 19 and 20. Renamed here: the field $L = \mathbb{Q}(a)$ is $\mathbb{L}$; the coordinate $L = qK$ of its Section 3 is $\tilde{K} = \mathfrak{q}K$; a rational exponent p/q is $\mathfrak{p}/\mathfrak{q}$. The Laguerre polynomials keep the name $L_m^{(\pm 1/2)}$. In the proof of Theorem 17.1, b is a positive integer and d_i are degrees of factors; neither is the depth.

17.1 An algebraic-degree bound

Theorem 17.1. *Let $a > 0$, $a \neq 1$ be algebraic of degree $D = [\mathbb{Q}(a) : \mathbb{Q}]$. Fix a defect r with*

$$\left\lfloor \frac{r}{2} \right\rfloor > 2D, \quad \text{equivalently} \quad r \geq 4D + 2.$$

Only finitely many physical cells have $p_{N,N-r,k}(a) = 0$ after excluding $K = 0$ for odd r . If a is irrational, no physical cell lies on that reflection line, so all coefficients on the fixed defect are nonzero for sufficiently large N .

[merge] Theorem 17.1 is implied by the other two fixed-defect theorems: for rational a ($D = 1$, $r \geq 6$) by source 20r's Theorem 16.1 ($r \geq 4$), and for irrational a by source 16's Theorem 18.1, which bounds the cancellation cells by $(2\lfloor r/2 \rfloor)^2 - 1$ on every defect $r \geq 2$, effectively. Source 20 says the first (after the proof below); source 16 calls the theorem “a prior sufficient algebraic-degree threshold” without naming it. The proof below, by field degrees and Siegel's theorem over $\mathbb{Q}(a)$, is a second route and is printed as such.

Proof (source 20). Put $\mathbb{L} = \mathbb{Q}(a)$ and $m = \lfloor r/2 \rfloor$. Express $k = (K + (2a - 1)N - 2ar + 1)/a$ in (15.6). Let $F_r(N, K)$ be this centered residual for even r , and its quotient by K for odd r . It is a polynomial over $\mathbb{L}$.

Assign weights two and one to N and K . Its top weighted part E_r has weight $2m$ and is given by (16.1)–(16.2). Indeed the highest-weight terms of the exponent in (15.6) are $-Kt/2 + cNt^2/24$, and the Hermite–Laguerre relations give those formulas. The assumption $a > 0$, $a \neq 1$ makes $c \neq 0$.

Schur's theorem makes each Laguerre polynomial in (16.1)–(16.2) irreducible of degree m over $\mathbb{Q}$ [8]. If θ is one of its roots, the field-degree inequality gives

$$[\mathbb{L}(\theta) : \mathbb{L}] \geq \frac{[\mathbb{Q}(\theta) : \mathbb{Q}]}{[\mathbb{L} : \mathbb{Q}]} = \frac{m}{D}.$$

The roots of $E_r(v, 1)$ have the form $-3/(2c\theta)$, so their degrees over $\mathbb{L}$ satisfy the same bound.

Factor $E_r(v, 1)$ over $\mathbb{L}$ into monic irreducibles $f_i(v)$ of degrees d_i . Each $d_i \geq \lceil m/D \rceil \geq 3$. Its corresponding homogeneous factor is

$$H_i(N, K) = K^{2d_i} f_i(N/K^2).$$

Purely transcendental extension and Gauss's lemma show that H_i is irreducible in $\mathbb{L}[N, K]$. The edge is squarefree because the Laguerre roots are simple.

Now factor F_r into irreducibles over $\mathbb{L}$. The top weighted part of each nonconstant factor is a nonempty product of distinct H_i . There is no nonconstant factor of degree zero in N , since the leading N coefficient of F_r is a nonzero constant. Thus each factor has degree at least $\lceil m/D \rceil$ in N . More explicitly, if its edge has weight $2n$, it contains a nonzero N^n term; weighted degree bounds its N degree by n , so that degree is exactly n .

Each such factor has distinct infinity branches. Setting $t = 1/K$, $N = v/t^2$ and multiplying by t^{2n} gives an analytic equation whose specialization at $t = 0$ has n simple roots. The implicit-function theorem gives n distinct normalization points, with local coordinate t , at which $K = t^{-1}$ has simple poles. Every component therefore has at least three geometric poles of K .

If an $\mathbb{L}$ -irreducible factor is not absolutely irreducible, its $\mathbb{L}$ -rational points lie in finite intersections of distinct Galois-conjugate absolute components. Otherwise, apart from finitely many singular points, they lift to the smooth projective normalization. Choose a fixed positive integer b with ba an algebraic integer. At every integer pair (N, k) ,

$$bK = (ba)(k - 2N + 2r) + b(N - 1) \in \mathcal{O}_{\mathbb{L}}.$$

The function bK has at least three poles. Siegel's theorem over $\mathbb{L}$ [9, Theorem 1.1] therefore gives only finitely many $\mathbb{L}$ -rational points where bK is integral. Taking the finite union over all components proves finiteness of cancellation cells off the discarded factor.

Finally, if a is irrational, the equation

$$a(k - 2N + 2r) + (N - 1) = 0$$

with integer variables forces $N = 1$ and $k = 2 - 2r$. This contradicts (15.2) for $r \geq 1$, proving the last assertion. $\square$

[merge] Source 20 displays the edge formulas again in its proof, under the label `ac:edges`; they are source 20r's (16.1)–(16.2) and are cited instead.

The degree threshold is sufficient, not claimed optimal. For rational $a \neq 1$, the companion theorem (Theorem 16.1) improves the threshold from six to four using two explicit genus-one quartics. The present argument does not classify every smaller defect at an arbitrary algebraic exponent.

17.2 Exact low-defect constants for rational exponents

Fix a reduced rational $a = \mathfrak{p}/\mathfrak{q}$ with $1/2 < a < 1$. Put

$$C = (\mathfrak{q} - \mathfrak{p})(2\mathfrak{p} + \mathfrak{q}) > 0, \quad \widetilde{K} = \mathfrak{q}K.$$

For $r = 2$ set $t_r = 3$, and for $r = 3$ set $t_r = 1$. Direct expansion of (15.6) gives the nonreflection zero equations

$$N = \frac{t_r \widetilde{K}^2 + \mathfrak{p} \widetilde{K} - 2\mathfrak{p}^2 r + \mathfrak{p}\mathfrak{q} - \mathfrak{q}^2}{C}, \quad (17.1)$$

$$k = \frac{\widetilde{K} + (2\mathfrak{p} - \mathfrak{q})N - 2\mathfrak{p}r + \mathfrak{q}}{\mathfrak{p}}. \quad (17.2)$$

For instance the centered residuals are $(3K^2 + U)/24$ and $-K(K^2 + U)/48$, with $U = aK + cN - 2a^2r + a - 1$.

Let S_r be the number of classes $\widetilde{K} \pmod{\mathfrak{p}C}$ making both expressions (17.1)–(17.2) integral. This is an explicit finite congruence count.

Theorem 17.2. *The number of physical nonreflection cancellation cells on defect $r \in \{2, 3\}$ through order X is*

$$\frac{2S_r}{\mathfrak{p}\sqrt{C}t_r} \sqrt{X} + O_{a,r}(1). \quad (17.3)$$

Moreover $S_r > 0$, and it is the product of the number of roots of

$$t_r \widetilde{K}^2 + \mathfrak{p} \widetilde{K} - 2\mathfrak{p}^2 r + \mathfrak{p}\mathfrak{q} - \mathfrak{q}^2 \equiv 0 \pmod{C}$$

and the number of roots of

$$t_r y^2 - y - 2 \equiv 0 \pmod{\mathfrak{p}}.$$

The latter polynomials factor as $(3y + 2)(y - 1)$ for $r = 2$, and $(y - 2)(y + 1)$ for $r = 3$.

Proof. Since $C \equiv \mathfrak{q}^2 \pmod{\mathfrak{p}}$ and $\gcd(\mathfrak{p}, \mathfrak{q}) = 1$, also $\gcd(\mathfrak{p}, C) = 1$. The first integrality condition is exactly the stated congruence modulo C . Modulo $\mathfrak{p}$, it gives

$$N \equiv t_r (\widetilde{K}/\mathfrak{q})^2 - 1.$$

Substituting in (17.2) makes the second integrality condition precisely $t_r y^2 - y - 2 \equiv 0$, where $y = \widetilde{K}/\mathfrak{q}$ modulo $\mathfrak{p}$. Multiplication by $\mathfrak{q}$ is a bijection modulo $\mathfrak{p}$, and the Chinese remainder theorem proves the product formula. The structural point $\widetilde{K} = \mathfrak{q}(r - 1)$, $N = r$, $k = 0$ proves that at least one class exists; it is used only to establish solvability, not as a physical cell.

As $|\widetilde{K}| \rightarrow \infty$, the first equation gives $N \sim t_r \widetilde{K}^2 / C$ and the second gives $k/N \rightarrow 2 - 1/a \in (0, 1)$. Thus both tails are physical, after deleting finitely many initial points. The constraint $N \leq X$ gives a real interval of length

$$2\sqrt{C/t_r} \sqrt{X} + O_{a,r}(1)$$

in the variable $\widetilde{K}$. Each admissible class modulo $\mathfrak{p}C$ contributes its interval length divided by $\mathfrak{p}C$, up to $O(1)$. Summing the S_r classes proves (17.3). The coordinate $\widetilde{K}$ determines (N, k) injectively. Omitting $\widetilde{K} = 0$ in the odd case affects at most a bounded term. $\square$

For example, $a = 3/4$ gives $C = 10$ and $S_2 = S_3 = 4$. Its two square-root constants are

$$\frac{8}{3\sqrt{30}}, \quad \frac{8}{3\sqrt{10}}.$$

At the endpoint $a = 1/2$, only the positive $\widetilde{K}$ tail is physical. The same congruence calculation has $C = 4$, $S_2 = S_3 = 2$, and halving (17.3) recovers the constants $1/\sqrt{3}$ and 1.

[merge] The endpoint constants count source 26's two families of Theorem 14.3: through order X they have $\sqrt{X/3} + O(1)$ and $\sqrt{X} + O(1)$ members. For $a = 3/4$ the batch-72A intake enumerated the two families to $X = 10^8$ and found 4,866 and 8,430 members, against 4,868.6 and 8,432.7 from (17.3); that check is not shipped.

Corollary 17.3. *For a fixed rational $1/2 < a = \mathfrak{p}/\mathfrak{q} < 1$ and fixed $R \geq 3$, the total number of cancellation cells on defects $0 \leq r \leq R$ through derivative order X is*

$$\frac{\lceil R/2 \rceil}{\mathfrak{p}} X + \frac{2}{\mathfrak{p}\sqrt{C}} \left(\frac{S_2}{\sqrt{3}} + S_3 \right) \sqrt{X} + O_{a,R}(1). \quad (17.4)$$

Proof. On each odd defect, the exact reflection family is

$$N = 1 + \mathfrak{p}h, \quad k = (2\mathfrak{p} - \mathfrak{q})h - 2(r - 1),$$

which contributes $X/\mathfrak{p} + O_{a,r}(1)$ physical cells. There are $\lceil R/2 \rceil$ odd defects in the window. Defect zero has none, and defect one has only reflection. Theorem 17.2 supplies the two nonreflection square-root terms. The rational fixed-defect theorem (Theorem 16.1) says every remaining fixed defect contributes only finitely many nonreflection exceptions; their finite sum is $O_{a,R}(1)$. All cells on different defects are disjoint. $\square$

[merge] Source 20 cites the companion (source 20r) here as “the accompanying files `rational_companion.pdf` and `rational_companion_source.zip`”. Neither file is shipped; the companion's text is Section 16, and its code and data are shipped under the prefix `20r-`. The corollary needs Theorem 16.1 and is new in source 20.

18 Irrational exponents: effective bounds (source 16)

[merge] This section is source 16, *Effective Bounds for Irrational Exponents: fixed-defect cancellations in derivatives of x^x* , from its Section 2 on; its Section 1 is Section 15 with Lemma 15.1, and its closing section is printed in Sections 19 and 20. Because it compares the residual at two conjugate values of the exponent, this section keeps source 16's notation $R_r(a; N, k) = R_{a,r,N}(k)$ and $K_a = K$ of (15.5). Renamed here: the leading line L_a is Λ_a ; in Section 18.3 the coefficient sum C_a is Σ_a , the constant A_a is Γ_a and the exponent $e = s_r$ is written s_r ; in the proof of Corollary 18.2, $d = 2u - k$, $q = 1/u$ and $t = d/u$ are δ , ϖ and τ . The letter b denotes a conjugate of a .

18.1 A uniform finite bound by conjugates

Let

$$s_r = 2\lfloor r/2 \rfloor.$$

Theorem 18.1. *For every positive irrational a and every fixed $r \geq 2$, at most $s_r^2 - 1$ physical cells satisfy $p_{N,N-r,k}(a) = 0$. There are no physical cancellations when $r = 1$.*

If a is transcendental, there are no cancellations on any fixed defect. If a is algebraic of degree $D > s_r$, there are no cancellations on defect r either.

Proof. The transcendental assertion follows immediately from Lemma 15.1. Suppose now that a is irrational algebraic, and choose a distinct conjugate b .

For even r , put $F_a = R_r(a; N, k)$. For odd $r \geq 3$, put $F_a = R_r(a; N, k)/K_a$. This quotient is polynomial, and its leading homogeneous part is

$$-\frac{\Lambda_a(N, k)^{r-1}}{2r!},$$

because the top homogeneous part of K_a is $-2\Lambda_a$. Thus in either parity F_a has total degree s_r and leading form a nonzero constant times $\Lambda_a^{s_r}$. The same holds at b .

At an integer cell the equation $K_a = 0$ would read

$$a(k - 2N + 2r) + (N - 1) = 0.$$

Irrationality forces $N = 1$ and $k = 2 - 2r$, contradicting (15.2). Hence a physical zero for odd r is a zero of F_a . Applying the embedding sending a to b shows it is also a zero of F_b .

The two leading lines are distinct, since

$$\det \begin{pmatrix} a - 1/2 & -a/2 \\ b - 1/2 & -b/2 \end{pmatrix} = \frac{b - a}{4} \neq 0. \quad (18.1)$$

A common nonconstant factor of F_a, F_b would have a leading homogeneous part dividing both $\Lambda_a^{s_r}$ and $\Lambda_b^{s_r}$, impossible. The two curves are therefore coprime. Their projective closures have no common point at infinity, again by (18.1). Bézout's theorem [11, I, Theorem 1.18] bounds their affine common points by s_r^2 , counting multiplicity.

The structural point $(N, k) = (r, 0)$ is always common: $R_r(a; r, 0) = [z^r]1 = 0$. In the odd case $K_a(r, 0) = r - 1 \neq 0$, so it survives division. This point lies outside (15.2); removing at least this one intersection leaves at most $s_r^2 - 1$ physical zeros.

For the degree restriction, the even case is Lemma 15.1. At a physical cell with odd r , K_a as a polynomial in a has nonzero leading coefficient

$$k - 2(N - r) \leq -(N - r) < 0.$$

The reduced coefficient polynomial consequently has exact degree $r - 1 = s_r$ in a , with rational coefficients. An algebraic number of degree greater than s_r cannot be its root. Finally, $R_1 = -K_a/2$, so the absence of irrational reflection cells settles $r = 1$. $\square$

In particular, every fixed defect is entirely nonzero beyond some derivative order. The bound is independent of the algebraic degree and does not invoke an integral-point finiteness theorem. It does not by itself control the union of defects growing with N .

[merge] The transcendental clause also follows from source 28's Corollary 10.1, which gives every transcendental exponent the full count $N(N+3)/2$ at every order; both rest on the coefficient polynomials not vanishing identically. For irrational a the theorem implies source 20's Theorem 17.1, effectively and on every defect $r \geq 2$.

18.2 Sharp bounds on defects two and three

Corollary 18.2. *For each fixed positive irrational exponent, there is at most one physical cancellation cell on defect two, and at most one on defect three. Both bounds are attained.*

Proof. Theorem 18.1 excludes degrees greater than two, so only quadratic irrational exponents need consideration. Put $u = N - r \geq 1$, $\delta = 2u - k$, and

$$\varpi = 1/u > 0, \quad \tau = \delta/u \in [1, 2].$$

For defect two, direct expansion gives

$$24R_2 = (3\delta^2 + k)a^2 - 6(u+1)\delta a + u(3u+5).$$

Normalize its positive constant term to one, writing the quadratic as $Pa^2 - Sa + 1$. Then

$$S = \frac{6\tau(1+\varpi)}{3+5\varpi}, \quad P = \frac{3\tau^2 + (2-\tau)\varpi}{3+5\varpi}.$$

A common irrational root forces two rational quadratics to be proportional, so their normalized pairs (S, P) agree. Along a fixed S , the function $\tau = S(3+5\varpi)/(6(1+\varpi))$ increases with ϖ . Direct differentiation gives

$$\left. \frac{dP}{d\varpi} \right|_S = -\frac{15\varpi\tau^2 + 5\varpi\tau - 6\varpi + 3\tau^2 + 3\tau - 6}{(\varpi+1)(5\varpi+3)^2} < 0.$$

The numerator equals $14\varpi > 0$ at $\tau = 1$ and increases with $\tau \geq 1$. Between two admissible values of ϖ , the increasing $\tau(\varpi)$ remains in $[1, 2]$. Thus P is strictly decreasing along that interval and determines ϖ uniquely. It then determines τ , hence u, δ, N, k .

For defect three, remove the nonvanishing irrational reflection factor K_a . The exact quadratic is

$$-48R_3/K_a = (\delta^2 + k)a^2 - 2(u+2)\delta a + u(u+3).$$

Its corresponding pair is

$$S = \frac{2\tau(1+2\varpi)}{1+3\varpi}, \quad P = \frac{\tau^2 + (2-\tau)\varpi}{1+3\varpi},$$

with $\tau = S(1+3\varpi)/(2(1+2\varpi))$ again increasing. Here

$$\left. \frac{dP}{d\varpi} \right|_S = -\frac{6\varpi\tau^2 + 3\varpi\tau - 4\varpi + \tau^2 + \tau - 2}{(2\varpi+1)(3\varpi+1)^2} < 0.$$

The numerator is 5ϖ at $\tau = 1$ and increases thereafter, proving uniqueness in the same way. The defect-two example in Section 18.4 attains one cell. At defect three, either $a = (3 \pm \sqrt{5})/2$ gives the physical cell $(N, k) = (4, 0)$, so that bound is also sharp. $\square$

[merge] In this proof P and S are the normalized coefficients of a quadratic, not a prefactor or an action, and R_2, R_3 abbreviate $R_2(a; N, k), R_3(a; N, k)$.

18.3 An explicit order bound and exact elimination

There is also a direct upper bound on the order of an algebraic cancellation. Use the compared residual F_a from Theorem 18.1, with $s_r \geq 2$, and write

$$F_a = c_r \Lambda_a^{s_r} + Q_a, \quad \deg Q_a \leq s_r - 1, \quad c_r = \begin{cases} 1/r!, & r \text{ even}, \\ -1/(2r!), & r \text{ odd}. \end{cases}$$

Let Σ_a be the sum of the absolute values of all coefficients of Q_a , and put $\Gamma_a = (\Sigma_a/|c_r|)^{1/s_r}$. Define Σ_b, Γ_b for any distinct conjugate b in the same way. At a physical zero, $N \geq 1$ and $0 \leq k \leq N$ imply

$$|Q_a(N, k)| \leq \Sigma_a N^{s_r-1}, \quad |\Lambda_a(N, k)| \leq \Gamma_a N^{1-1/s_r},$$

and similarly for b . Since $b\Lambda_a - a\Lambda_b = (a-b)N/2$, every physical zero satisfies

$$\boxed{N \leq \left(\frac{2(|b|\Gamma_a + |a|\Gamma_b)}{|a-b|} \right)^{s_r}}. \quad (18.2)$$

All quantities are algebraic and computable from exact algebraic input. Rational upper bounds for the numerator and a positive rational lower bound for $|a-b|$ give an explicit finite search rectangle. The case $r = 1$ has no zeros and needs no such bound.

Suppose a is specified by an exact irreducible minimal polynomial $P(A) \in \mathbb{Q}[A]$ of degree $D \geq 2$, together with an embedding selecting a positive real root. Fix r . Reduce the raw residual modulo P :

$$R_r(A; N, k) \equiv \sum_{j=0}^{D-1} A^j f_j(N, k) \pmod{P(A)}, \quad f_j \in \mathbb{Q}[N, k]. \quad (18.3)$$

For integer N, k , cancellation is equivalent to $f_j(N, k) = 0$ for every j , by linear independence of $1, a, \dots, a^{D-1}$ over $\mathbb{Q}$.

The common complex variety of the f_j is contained in the intersection of the raw conjugate curves $R_r(a; N, k) = R_r(b; N, k) = 0$. The leading-line argument of (18.1), applied before any odd division, makes this intersection finite. Thus the rational polynomial system (18.3) is zero-dimensional or empty.

Exact Gröbner elimination or resultants therefore compute a finite candidate set. Concretely, a nonzero univariate elimination polynomial in N supplies all possible integer N by rational-root extraction. At each such value, the remaining polynomials in k supply all possible integer k . Check every candidate in all f_j and apply (15.2). Equivalently, univariate elimination polynomials in both coordinates give explicit rational Cauchy bounds, followed by a finite exact integer enumeration. A unit ideal means there are no candidates.

This is a terminating exact procedure for the complete physical zero list. One plus its largest derivative order is an explicit eventual threshold; if the list is empty, the entire physical range is already nonzero. The statement does not promise a small computational cost. It requires exact algebraic input, and does not provide an algorithm to recognize transcendence from arbitrary numerical data.

18.4 A complete quadratic example

Take either

$$a = 1 \pm \frac{1}{\sqrt{3}} > 0, \quad P(A) = 3A^2 - 6A + 2,$$

and fix $r = 2$. At $(N, k) = (3, 0)$,

$$R_2(a; 3, 0) = a^2/2 - a + 1/3 = 0.$$

Thus an irrational exponent can have a genuine physical cancellation.

Reducing the full residual modulo P and computing a rational Gröbner basis gives the equivalent equations

$$(N - 2)(N - 3)(9N^2 - N - 2) = 0, \quad (18.4)$$

$$k = -\frac{3}{4}N^3 + \frac{37}{12}N^2 - \frac{7}{6}N - 4. \quad (18.5)$$

The only integer roots of (18.4) are $N = 2, 3$, since the remaining quadratic has nonsquare discriminant 73. Both yield $k = 0$. The point $(2, 0)$ is structural and lies outside $N \geq 3$. Therefore, for each of the two exponents, the complete physical defect-two zero list is

$$\boxed{\{(N, k) = (3, 0)\}}.$$

Every coefficient on that defect is nonzero at every order $N \geq 4$. The accompanying exact checker verifies both directions of the ideal equivalence in (18.4)–(18.5), so this is a complete elimination rather than a bounded numerical search.

There is no uniform derivative-order cutoff as the irrational exponent varies, even at defect two. For an integer $u \geq 1$, direct substitution gives

$$R_2(a; u + 2, 0) = \frac{u}{24}(12ua^2 - 12(u + 1)a + 3u + 5).$$

Its two roots are

$$a = \frac{u + 1}{2u} \pm \frac{\sqrt{3(u + 3)}}{6u}.$$

Both are positive, and they are quadratic irrational whenever $3(u + 3)$ is nonsquare. Taking $u = 3h^2$ with integers $h \geq 1$ supplies such exponents with unbounded cancellation order $N = u + 2$. Thus the cell-count bound is uniform, while an eventual order threshold genuinely depends on the exponent.

[merge] In this example $P(A)$ is a minimal polynomial in the variable A , not the series A_a . The two exponents $1 \pm 1/\sqrt{3}$ are the roots of source 28's $p_{3,1,0}(\alpha) = 3\alpha^2 - 6\alpha + 2$; the batch-72A intake also checked that $p_{4,1,0}(\alpha) = 2(2\alpha - 3)(\alpha^2 - 3\alpha + 1)$, whose quadratic factor has the roots $(3 \pm \sqrt{5})/2$ of the defect-three example. “The accompanying exact checker” is shipped as `code/16-irrational-verify.py`.

Scope, verification and provenance

19 Source positions, scope and open questions

Each source closes with a statement of its position relative to the host report and the literature, and of what it does not claim. They are printed here, source by source, in the sources’ words; Section 19.8 collects the open questions.

19.1 Integer exponents (source 28)

Question 8 of the inspected ProveIt source [1] explicitly asks for extension to x^{x^r} , $r \geq 3$, and cautions that the coefficient generating function and symmetry center must be rederived. This

paper supplies both and proves the leading-support theorem for every fixed positive integer exponent. For $a = 1$ and $a = 2$, the live OEIS entries [2, 3] concern the same derivative-count convention; the exact coefficient triangles and their identities have classical antecedents [5]. No claim of exhaustive worldwide priority is made.

The proof is conventional mathematics, not a proof-assistant formalization or an externally refereed publication. Its asymptotic constants and thresholds may depend on the fixed exponent. The exact support formula for general a , a full classification of deeper cancellations, and any theorem with a growing with N remain separate questions. Corollary 10.1 settles the generic real parameter by an elementary polynomial argument, but it does not settle the exceptional noninteger algebraic parameters.

[merge] Source 28’s bibliography calls the host’s Section 3 “Part II”; the host numbers it Part 3. Source 28 does not mention the host’s Question 6, which its case $a = 2$ answers.

19.2 The exponent $a = 1$ (source 31)

The inspected repository source [1], Part I, states only $a(N) = \Theta(N^2)$ and isolates $a(N) \sim N^2/2$ as a weaker target than its exact zero conjecture. The live OEIS entry [2], checked on 1 October 2026, still attributes the asymptotic conjecture to V. Kotešovec, dated 5 October 2017. Jeong’s recent primary treatment [5] develops coefficient identities and recurrences; no zero-density statement was located there. This is a bounded source comparison, not an exhaustive claim of worldwide priority.

The proof is a conventional analytic argument, not a Lean formalization or an externally refereed publication. It does not identify every zero of A008296, prove the proposed exact rational generating function for A293239, or remove the known sporadic cancellation $b(8, 5) = 0$.

[merge] “Part I” is the host’s Part 2 (its Section 2). The statements it describes are the host’s Theorem 2.16 and equation (2.35), `xx:eq:asympeq`.

19.3 The exponent $a = 2$ (source 34)

The proof supplies explicit contours in every simple favored-saddle regime. It does not infer dominance from a critical-point equation, a genericity assumption, or a proposed homology decomposition. The contour-switch and turning sets are removed only for an area argument; no claim is made about every coefficient on them.

A bounded primary-source check on 1 October 2026 confirmed that the current OEIS entry [3] still labels its generating function and exact formula conjectural. The inspected repository article still lists the leading asymptotic as Question 6 in its depth-coordinate research questions; its latest file commit remains the one cited below. Corollary 13.1 answers that leading-asymptotic question, with the exact support problem still unresolved.

The complete argument received independent mathematical review, including a fresh review of the full-cone conclusion. It has not been formalized in Lean or externally refereed, and no exhaustive priority claim is made. The leading asymptotic is proved here; the stronger assertion that the known reflection holes are the only cancellations, and the resulting exact term-count formula or generating function, remain unresolved.

[merge] Source 34 writes “Theorem” for its main statement, printed here as Corollary 13.1. “The one cited below” is the pin `ca81647a9` in the bibliography entry [1]; it was the latest commit to the host’s article when source 34 was written, and it is kept as provenance, not as a statement about the host’s current text. The “independent mathematical review” is the source’s own statement and was not repeated by the repository.

19.4 The half exponent (source 26)

The inspected ProveIt derivative-support report [1] explicitly asks for extensions beyond the square exponent. This paper treats the specific noninteger algebraic parameter $a = 1/2$. A bounded repository and primary-source search located no matching half-exponent derivative-count theorem; this is not an exhaustive priority claim. We do not assign this sequence an OEIS identifier.

The proof is conventional analysis, not a proof-assistant formalization or an externally refereed publication. A complete all-defect zero classification and a global quantitative asymptotic remainder remain open. The negative effective exponent in (14.7) explains why the integer-exponent horizontal proof cannot simply be copied; the positive terminal derivative (14.9) supplies the required separate argument.

[merge] “The integer-exponent horizontal proof” is source 28’s Section 5, which source 26 does not cite.

19.5 Rational exponents (source 20r)

The inspected ProveIt manuscript asks about derivative supports for further exponents [1]; this note is an arithmetic continuation in that direction. Schur irreducibility, the orthogonal-polynomial identities and Siegel finiteness are classical inputs. No exhaustive priority claim is made. The full finite exceptional lists, effective thresholds, the exponent $a = 1$ fixed-defect integer-root problem, and global support-density theorems for general noninteger rational exponents are separate questions. The proof is conventional mathematics, not proof-assistant formalization or an externally refereed publication.

19.6 Algebraic exponents (source 20)

The rational fixed-defect note used in Corollary 17.3 is included as a companion PDF and source package, so its explicit quartic argument and certificates are available with this report. No numerical effective threshold, uniform growing-defect assertion, or complete finite exceptional list is claimed. The exceptional exponent $a = 1$ remains a separate persistent-zero-line problem. Classical inputs are attributed below, and no exhaustive priority claim or formal verification is asserted.

[merge] The companion PDF and source package are not shipped. The companion’s text is Section 16 of this report, and its verifier, certificates and replay are shipped as the four 20r- files (Section 20).

19.7 Irrational exponents (source 16)

The result continues the derivative-support program in the inspected ProveIt source [1]. Bézout’s theorem and exact polynomial elimination are classical inputs. This theorem strengthens a prior sufficient algebraic-degree threshold to every fixed defect, with an explicit uniform cell bound and effective elimination for algebraic inputs. Rational exponents are excluded from the conjugate argument and retain their separate arithmetic behavior. Complete zero classification when defects grow, practical complexity bounds for elimination, and global support asymptotics for arbitrary algebraic exponents are not asserted. No exhaustive priority claim, proof-assistant formalization or external-referee approval is made.

[merge] “A prior sufficient algebraic-degree threshold” is source 20’s Theorem 17.1.

19.8 The open questions, collected

[merge] This list is written for the merge. Every item is stated as open by at least one source, named after it; none is answered by another source.

1. *Exact supports.* Classify every cancellation: for $a = 1$, every zero of the Lehmer–Comtet triangle A008296, which would settle the host’s Conjecture 2.9 and the proposed rational generating function of A293239 (source 31); for $a = 2$, whether the reflection holes are the only cancellations, and with it the exact term-count formula and generating function of A290268 (source 34); for every integer a , the exact support formula and the deeper cancellations (source 28); for $a = 1/2$, every defect (source 26).
2. *Quantitative remainders.* A power-saving remainder or an effective order threshold in Theorems 3.1 and 14.1 (sources 28, 31, 34, 26); effective thresholds and complete finite exceptional lists on fixed defects (sources 20r, 20).
3. *Growing parameters.* Uniformity when a grows with N (source 28); support density, or a zero classification, when the defect grows with N (sources 20r, 20, 16).
4. *Non-integer exponents globally.* Global support asymptotics for general non-integer rational (source 20r) and arbitrary algebraic exponents (source 16), beyond the half exponent; the exceptional algebraic parameters of Corollary 10.1 (source 28); the defects below $4D + 2$ at an algebraic exponent of degree D (source 20), and whether that threshold is optimal (source 20); singular quartics at irrational algebraic parameters (source 20r).
5. *The exponent $a = 1$ on fixed defects.* The integer-root problem of the centered residual at $a = 1$, where roots give persistent zero lines (sources 20r, 20).
6. *Computation.* Practical complexity bounds for the elimination of Section 18.3, and recognition of transcendence from numerical data (source 16).

[merge] *A finite data point for the host’s Question 8 and Research question 3.68.* The batch-72A intake scanned the exact recurrence (10.1) for $a = 1, \dots, 5$ and $N \leq 70$. For $a = 2, 3, 4, 5$ every zero is structural ($d = 0$) or a reflection zero on $q = ad$ with $(a - 1)k + d$ even (144, 60, 36 and 18 of them); there is no other zero. For $a = 1$ there are 578 reflection zeros and 63 others, which are the host’s sporadic zero $b(8, 5) = 0$ repeated along the line $(N, \ell, k) = (k + 8, k + 5, k)$, $0 \leq k \leq 62$. This is a finite observation, not a theorem; the scan is not shipped.

20 Verification, files and provenance

20.1 The delivered checks

Each source’s verification paragraph is printed here, followed by the names under which its files are shipped. Every program writes its output beside itself under its delivered name (the README gives rerun instructions that leave the shipped files untouched). None of these finite checks replaces an analytic proof.

Source 28. The standard-library verifier `verify_general_coefficients.py` checks 631 exact identities for $a = 1, \dots, 8$, including 104 zero cases, 187 strictly positive bulk cases, and 252 depth-one/two sign cases. The output is recorded in `general_coefficient_checks.json`. These finite checks supplement the analytic proof; they do not replace its global contour selection or establish exact noncancellation at arbitrary depth.

[merge] Shipped as `code/28-all-integer-verify_general_coefficients.py` and `data/28-all-integer-general_coefficient_checks.json`.

Source 31. The accompanying standard-library program `verify_coefficients.py` independently checks 510 rational coefficient identities in (12.17), including seven zero cases. It also compares the exact derivative recurrence with the Lehmer–Comtet recurrence through order 200. For example $a(100) = 5026$ and $a(200) = 20051$. The optional `check_saddle_normalization.py` checks the Gaussian factors numerically in real, low-complex, and horizontal regimes; these computations are diagnostic and are not substitutes for the proof.

[merge] Shipped as `code/31-a293239-verify_coefficients.py` with `data/31-a293239-coefficient_checks.json`, and `code/31-a293239-check_saddle_normalization.py` (needs `mpmath`) with `data/31-a293239-saddle_normalization.json`.

Source 34. The exact transformed coefficient identity was independently replayed against the source w -formula for 432 integer triples, including 16 zero cases. A separate rational certificate verifies the elementary trigonometric inequalities used to show that the phase curvature is nontrivial. An independent exact derivative-recurrence regression through $N = 200$ checks the support envelope, bulk, and reflection holes; for example $a(100) = 5101$ and $a(200) = 20201$. These finite checks detect normalization mistakes and do not prove the exact noncancellation conjecture; the asymptotic theorem follows from the written analytic and counting proofs.

[merge] Shipped as `code/34-a290268-verify_tanh_identity.py`, `-verify_curvature_seed.py` and `-verify_derivative_counts.py`, with `data/34-a290268-tanh_identity_checks.json`, `-curvature_seed.json` and `-derivative_counts.json`. The last records its own running time, so a rerun never reproduces it byte for byte.

Source 26. The verifier `verify_half_coefficients.py` uses exact rational arithmetic to check 455 reciprocal-transform identities and 33,523 complete coefficient identities on defects one through three. An independent integer recurrence scans all derivative rows through $N = 150$. In that finite range, the two families above account for all 16 observed nonreflection zeros. That observation is not used to infer a global classification. For orientation, $A(50) = 1324$, $A(100) = 5149$, and $A(150) = 11474$.

[merge] Shipped as `code/26-half-exponent-verify_half_coefficients.py` and `data/26-half-exponent-half_coefficient_checks.json`. “The two families above” are those of Theorem 14.3.

Source 20r. The supplied symbolic verifier derives the residual directly from the A_a/B_a power series for arbitrary symbolic a through $r = 6$. It checks the exact reflection factor, weighted degree, both Laguerre constants, the fourth and fifth residual formulas, the two discriminant identities, and the modular certificates. A separate standard-library verifier derives the two discriminant identities using the closed depressed-quartic formula, verifies both modular remainders, and checks 19,712 direct differentiation versus power-series identities through $N = 28$. These include 4,984 centered fourth/fifth-defect identities and 103 physical reflection checks. These are finite exact identity checks. The unrestricted fixed- r argument uses the written generating-function proof, Schur’s theorem, the normalization branches and Siegel’s theorem.

[merge] Shipped as `code/20r-rational-verify.py` (needs `SymPy`) with `data/20r-rational-certificates.json`, and `code/20r-rational-replay_independent.py` with `data/20r-rational-replay_independent.json`. The replay reads `certificates.json` from its own directory.

Source 20. A separate standard-library checker verifies the congruence-product formula and direct integer parametrization in 102 parameter/defect cases with denominator at most 18, including the half-exponent endpoint and the example $3/4$. The algebraic-degree theorem is proved uniformly by field degrees, factorization and Siegel’s theorem, not by a finite scan.

[merge] Shipped as `code/20-algebraic-replay_low_counts.py` and `data/20-algebraic-replay_low_counts.json`.

Source 16. The symbolic verifier derives the residual from the defining power series and checks the degree statements, parity reduction, and structural point through $r = 6$. A separate

exact checker verifies both low-defect quadratic formulas and the fixed- S derivative identities. The first checker also reproduces the quadratic minimal-polynomial reduction, the Gröbner basis, and the exact integer filtering. These finite identities supplement the all- r leading-form and conjugate proof.

[merge] Shipped as `code/16-irrational-verify.py` with `data/16-irrational-checks.json`, and `code/16-irrational-check_low_defect_injectivity.py`, which prints its result and writes no file. Both need SymPy.

[merge] At placement every shipped program was rerun on a copy; each output equalled the delivered one apart from line endings and the timing field of `34-a290268-derivative_counts.json`.

20.2 Provenance

This report was built from seven manuscripts of batch 72A (Table 1): base 28, members 31, 34, 26, 20, 20r and 16. All arrived in commit 1512ef835, inside six archives; 20r is the folder `Rational_Exponent_Fixed_Defects` of the zip `rational_companion_source.zip` in archive 20 (`Algebraic_Exponents_and_Cancellation_Frequencies/`). The placement commit d292c6765 staged source 28’s `article.tex` and `README.md` unprefixed and the code and data of all seven with the prefixes `28-all-integer-`, `31-a293239-`, `34-a290268-`, `26-half-exponent-`, `20-algebraic-`, `20r-rational-` and `16-irrational-`; this text replaces both staged files. Sources 28, 31, 34, 26, 20r and 16 pin the host at `ca81647a9`; source 20 names no commit.

What each contributed: source 28 the general theorem for integer exponents, its proof, the generic-parameter corollary and the depth-one and depth-two classification (Sections 3–11); source 31 the case $a = 1$ with its explicit Lehmer–Comtet analysis (Section 12); source 34 the case $a = 2$ with its `tanh` sections and curvature certificate (Section 13); source 26 the half exponent and its two families (Section 14); source 20r the centered identity and the rational fixed-defect theorem (Sections 15, 16); source 20 the algebraic-degree theorem and the exact low-defect frequencies (Section 17); source 16 the degree lemma and the effective irrational bounds (Sections 15, 18). Where the merge had to choose is recorded in Section 1.2.

Not shipped: the manuscripts, READMEs and PDFs of sources 31, 34, 26, 20 and 16 (with 34’s `tanh_sections.tex`, printed in Section 13); source 28’s delivered PDF; source 20r’s `article.tex`, `article.pdf` and `README.md`; archive 20’s `rational_companion.pdf` and the zip itself; and the seven copies of a Linux build script, `build_local.sh`, one in each of the seven packages. They survive in commit 1512ef835.

Every source carries the author line “Research note prepared for Vladimir Reshetnikov with OpenAI”. Their READMEs state that each proof received independent mathematical review and that every rendered page was inspected; those statements are the sources’ own. This report is AI-assisted and unrefereed, and none of its results is formalized.

References

- [1] V. Reshetnikov, repository owner, *Term counts in the derivatives of power towers*, ProveIt research report `power-tower-derivative-term-counts`: Part 2 (the sources’ “Part I”) on x^x , Part 3 (the sources’ “Part II”) on x^{x^2} , including its depth-coordinate Questions 6 and 8. File commit `ca81647a9f679e96d49ffdbc54b0c0ab13d10a36`, inspected 1 October 2026.

Pinned source: `power-tower-derivative-term-counts/article.tex`.

- [2] V. Reshetnikov and contributors, *A293239: Number of terms in the fully expanded n th derivative of x^x* , The On-Line Encyclopedia of Integer Sequences. Asymptotic conjecture attributed to V. Kotešovec. Checked 1 October 2026. <https://oeis.org/A293239>.
- [3] V. Reshetnikov and contributors, *A290268: Number of terms in the fully expanded n th derivative of x^{x^2}* , OEIS, inspected 1 October 2026. <https://oeis.org/A290268>.
- [4] N. J. A. Sloane and contributors, *A008296: Triangle of Lehmer–Comtet numbers of the first kind*, OEIS. <https://oeis.org/A008296>.
- [5] S. Jeong, *On a Question of Lehmer concerning the Comtet Numbers*, arXiv:2607.26613v1, 2026. <https://arxiv.org/html/2607.26613v1>.
- [6] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, §4.22 and §4.36 (trigonometric and hyperbolic infinite products and partial fractions), equations 4.22.3 and 4.36.3. <https://dlmf.nist.gov/4.22.E3>; <https://dlmf.nist.gov/4.36.E3>.
- [7] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, equations 18.7.11, 18.7.19 and 18.7.20. <https://dlmf.nist.gov/18.7>.
- [8] S. Laishram and T. N. Shorey, *Irreducibility of generalized Hermite–Laguerre polynomials III*, arXiv:1306.0736v2 (2016), pp. 1–2, recording Schur’s 1929 and 1931 results on half-integer Laguerre irreducibility. <https://arxiv.org/abs/1306.0736>.
- [9] A. Levin, *Siegel’s theorem and the Shafarevich conjecture*, Journal de Théorie des Nombres de Bordeaux **24** (2012), 705–727, Theorem 1.1. <https://jtnb.centre-mersenne.org/item/10.5802/jtnb.818.pdf>.
- [10] S. Lang, *Integral points on curves*, Publications Mathématiques de l’IHÉS **6** (1960), 27–43, Theorem 4. https://www.numdam.org/article/PMIHES_1960__6__27_0.pdf.
- [11] J. S. Milne, *Elliptic Curves*, BookSurge, 2006, Chapter I, Theorem 1.18 (Bézout), p. 13. <https://www.jmilne.org/math/Books/ectext6.pdf>.

[merge] The bibliography merges the sources’ own. Each source’s entry for the host became [1]; source 34’s two DLMF entries and sources 28, 31 and 26’s single one became [6]; source 20’s entry for its companion became references to Section 16.